-->

Hematopoietic Cell Transplantation for Autoimmune Diseases and Miscellaneous Indications

Number: 0606

Table Of Contents

Policy
Applicable CPT / HCPCS / ICD-10 Codes
Background
References

Policy

Scope of Policy

This Clinical Policy Bulletin addresses hematopoietic cell transplantation for autoimmune diseases and miscellaneous indications.

I. **Medical Necessity**

Aetna considers autologous hematopoietic cell transplantation medically necessary for the treatment of adults (18 to 69 years of age) with rapidly progressive scleroderma (systemic sclerosis) at risk of organ failure when *either* of the following is met:

A. Active interstitial lung disease (as determined by broncho-alveolar cell composition or ground-glass opacities on computed tomography of the chest) plus either a forced vital capacity (FVC) or a diffusing capacity of the lung for carbon monoxide (DLCO) of less than 70 % of the predicted value; *or*
B. Previous scleroderma-related renal disease; *and*
C. The member meets the transplanting institutions' selection criteria, or in the absence of such criteria, the member does not have *any* of the following exclusion criteria:

1. Poor renal function (creatinine clearance of less than 40 ml/min); *or*
2. Severe lung disease (diffusing capacity of the lung for carbon monoxide (DLCO) of less than 40 % of the predicted value; or forced vital capacity (FVC) of less than 45 % of the predicted value); *or*
3. Pulmonary arterial hypertension; *or*
4. Poor lung function (left ventricular ejection fraction (LVEF) of less than 50%).

II. **Experimental, Investigational, or Unproven**

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

A. Aetna considers hematopoietic cell transplantation (autologous or allogeneic) experimental, investigational, or unproven for *any* of the following because the effectiveness of these approaches for these autoimmune diseases/conditions has not been established:

1. Autoimmune diseases (not an all-inclusive list):

    - Anti-phospholipid syndrome
    - Autoimmune cytopenia (e.g., autoimmune hemolytic anemia, Evans syndrome, and idiopathic thrombocytopenic purpura)
    - Autoimmune diseases-induced cirrhosis
    - Autoimmune retinopathy
    - Behcet's disease with myelodysplastic syndrome
    - Celiac disease
    - Chronic inflammatory demyelinating polyradiculopathy
    - Cogan syndrome
    - Crohn's disease
    - Dermatomyositis
    - Diabetes mellitus type I
    - Inflammatory bowel disease
    - Juvenile idiopathic arthritis and associated lung disease
    - Juvenile rheumatoid arthritis
    - Lupus nephritis
    - Multiple sclerosis
    - Myasthenia gravis
    - Neuromyelitis optica / neuromyelitis optica spectrum disorder
    - Pemphigus
    - Polymyositis
    - Rheumatoid arthritis
    - Sporadic late-onset nemaline myopathy
    - Stiff-person syndrome / Stiff-person spectrum disorder
    - Systemic lupus erythematosus
    - Systemic vasculitis
    - Ulcerative colitis.

2. Miscellaneous conditions (not an all-inclusive list):

    - Age-related macular degeneration
    - Amyotrophic lateral sclerosis
    - Essential thrombocythemia
    - Polycythemia vera
    - Recessive dystrophic epidermolysis bullosa
    - Retinitis pigmentosa
    - Thrombotic thrombocytopenic purpura.

B. Aetna considers the following interventions experimental, investigational, or unproven:

- The use of mesenchymal stem cell transplantation for the treatment of autoimmune diseases (including ankylosing spondylitis and Sjogren's syndrome) and selected indications listed in policy statement II.B;
- Administration (intra-arterial, intra-thecal, and intravenous) of autologous mesenchymal stem cells for the treatment of multiple system atrophy;
- Autologous umbilical cord blood stem cells for the treatment of autism;
- HLA-haploidentical hematopoietic stem cell transplantation for chronic granulomatous disease
- Pluripotent stem cell-based retinal organoids for restoration of blindness
- Umbilical cord blood transplantation in children with neuronal ceroid lipofuscinosis.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

CPT codes not covered for indications listed in the CPB:

***Administration (intra-arterial, intra-thecal, and intravenous) of autologous mesenchymal stem cells* - no specific code**

38204	Management of recipient hematopoietic progenitor cell donor search and cell acquisition
38205	Blood-derived hematopoietic progenitor cell harvesting for transplantation, per collection; allogeneic
38206	Blood-derived hematopoietic progenitor cell harvesting for transplantation, per collection; autologous
38207 - 38215	Bone marrow or stem cell services/procedures
38230 - 38232	Bone marrow harvesting for transplantation; allogeneic or autologous
38240	Hematopoietic progenitor cell (HPC); allogeneic transplantation per donor
38241	autologous transplantation

Other CPT codes related to the CPB:

86813	HLA typing; A, B or C, multiple antigens
86817	DR/DQ, multiple antigens
86821	lymphocyte culture, mixed (MLC)
86920 - 86923	Compatibility test each unit

HCPCS codes not covered for indications listed in the CPB:

S2140	Cord blood harvesting for transplantation, allogeneic
S2142	Cord blood-derived stem-cell transplantation, allogeneic
S2150	Bone marrow or blood-derived stem cells (peripheral or umbilical), allogeneic or autologous, harvesting, transplantation, and related complications; including: pheresis and cell preparation/storage; marrow ablative therapy; drugs, supplies, hospitalization with outpatient follow-up; medical/surgical, diagnostic, emergency, and rehabilitative services; and the number of days of pre- and post-transplant care in the global definition

ICD-10 codes covered for indications listed in the CPB:

M34.0 - M34.9	Systemic sclerosis [scleroderma]

ICD-10 codes not covered for indications listed in the CPB:

D45	Polycythemia vera

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Code	Code Description
D46.9	Myelodysplastic syndrome, unspecified [Behcet's disease with myelodysplastic syndrome]
D47.3	Essential (hemorrhagic) thrombocythemia
D59.0 - D59.19	Autoimmune hemolytic anemia
D68.61	Antiphospholipid syndrome
D69.3	Immune thrombocytopenic purpura
D69.41	Evan's syndrome
D71	Functional disorders of polymorphonuclear neutrophils [chronic granulomatous disease]
E08.00 - E13.9	Diabetes mellitus
E75.4	Neuronal ceroid lipofuscinosis
F84.0 - F84.9	Pervasive developmental disorders
G12.21	Amyotrophic lateral sclerosis
G25.82	Stiff-man syndrome
G35	Multiple sclerosis
G36.0	Neuromyelitis optica [Devic]
G61.81	Chronic inflammatory demyelinating polyneuritis
G70.00 – G70.01	Myasthenia gravis
G72.49	Other inflammatory and immune myopathies, not elsewhere classified
G90.3	Multi-system degeneration of the autonomic nervous system
H16.321 – H16.329	Diffuse interstitial keratitis
H35.30	Unspecified macular degeneration (age-related)
H35.52	Pigmentary retinal dystrophy
H35.89	Other specified retinal disorders [autoimmune retinopathy]
H54.0X33 - H54.8	Blindness and low vision
J98.4	Other disorders of lung [associated lung disease]
K50.00 - K50.919	Crohn's disease [regional enteritis]
K51.00 - K51.919	Ulcerative colitis
K75.4	Autoimmune hepatitis [autoimmune disease induced cirrhosis]
K90.0	Celiac disease
L10.9	Pemphigus, unspecified
M05.00 - M06.9	Rheumatoid arthritis
M08.90	Juvenile arthritis, unspecified, unspecified site [juvenile idiopathic arthritis]
M31.10 - M31.19	Thrombotic microangiopathy
M32.10 - M32.9	Systemic lupus erythematosus (SLE)
M33.00 - M33.99	Dermatomyositis
M35.00 – M35.09	Sjogren syndrome
M35.2	Behcet's disease [Behcet's disease with myelodysplastic syndrome]
M36.0	Dermato(poly)myositis in neoplastic disease
M45.0 - M45.9	Ankylosing spondylitis
Q81.0 - Q81.9	Epidermolysis bullosa

Background

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Autoimmune diseases (ADs) include a heterogeneous group of immune-mediated disorders that are responsive to suppression or modulation of the immune system. Some common ADs are multiple sclerosis (MS), rheumatoid arthritis (RA), and systemic lupus erythematosus (SLE). The prevalence of ADs in the United States is estimated to be approximately 2 %. In particular, MS afflicts about 350,000 people in the United States, while RA and SLE affects 0.5 to 1.0 % and 0.05 % of Americans, respectively. These diseases are often characterized by chronic, painful and debilitating courses that warrant aggressive therapy. For some patients with severe, relapsing/refractory cases, conventional therapy may not be satisfactory.

High-dose chemotherapy (HDC) and bone marrow/peripheral stem cell transplantation (autologous or allogeneic) has been studied for the treatment of severe ADs. The notion of employing HDC and bone marrow/peripheral stem cell transplant to treat AD is based on encouraging results in experimental animals and from serendipitous reports of patients with both ADs and malignancies who were allotransplanted for the latter. High- dose chemotherapy and bone marrow/peripheral stem cell transplant has been tried for patients with ADs who are refractory to standard therapies and are at high-risk of subsequent morbidity and mortality. Multiple sclerosis, RA, SLE, and systemic sclerosis are the ADs that have been most commonly treated by this procedure. Although early findings are promising, the number of patients treated is limited, and only short-term follow-up is available. Furthermore, the mechanism of improvement or stabilization is unclear, and the procedure has the potential for life-threatening toxicity.

Systemic Sclerosis (Scleroderma)

A review (Bingham et al, 2000) stated that autologous stem cell transplantation is starting to be examined as a potential therapy for severe, refractory ADs including neurological, rheumatological, and hematological diagnoses. Increasing numbers of cases are now reported in the scientific literature. Data from all transplanted patients are being collated in a centralized register by the European Group for Blood and Marrow Transplantation and the European League against Rheumatism to ensure effective assessment of the safety and effectiveness of this promising procedure. Thus far, results have been encouraging; however, they need to be confirmed by well-designed randomized, controlled studies in view of the well-known difficulty of judging objectively the effect of a treatment in patients with these diseases. Optimization of mobilization, conditioning regimen, as well as graft manipulation is needed to maximize effectiveness without increasing mortality and morbidity. The use of maintenance therapy after autologous stem cell transplantation to prevent relapse needs to be investigated. Furst (2000) noted that stem cell transplantation for the treatment of systemic sclerosis is showing some apparent effectiveness, but its use is only in the pilot stages.

Witkowska et al (2014) stated that autoimmune connective tissue diseases (ACTDs) are heterogeneous disorders associated with different manifestations, clinical course of disease and prognosis among patients. Although recent advances in understanding the pathogenesis have led to major progress in target-oriented therapy, they still remain incurable. Novel biological drugs, cellular therapy and HSCT are real hopes for treatment development in the future. The concept of both autologous and allogeneic HSCT in children with autoimmune diseases is developing energetically since 1996, when the first HSCT was performed. Nowadays, after 17 years of clinical experience, both types of HSCT remain attractive and powerful salvage methods of treatment. However, there are still many doubts and unclear issues, which need further investigation. These investigators provided an overview of the knowledge concerning actual data on HSCT in a pediatric group of patients with different ACTDs, focused on juvenile idiopathic arthritis (JIA), SLE and systemic sclerosis.

Cipriani et al (2015) stated that the presence of autoimmune diseases, including systemic sclerosis (SSc), suggest failure of the normal immune regulatory processes leading to activation and expansion of auto-reactive effector immune cells. Recently, stem cell transplantation emerged as a novel rescue therapy for a variety of refractory autoimmune diseases. The therapeutic strategy involves the ablation of the aberrant self-reactive immune cells by chemotherapy and the regeneration of a new self-tolerant immune system formed by the transplanted stem cells. In the last few years, thousands of patients worldwide have received HSCT, mostly autologous, as treatment for severe irreversible autoimmune diseases, with promising results. These investigators reviewed the results of published small series of SSc patients treated with allogeneic and autologous HSCT, as well as 3 randomized trials, exploring the safety and effectiveness of autologous HSCT in SSc. The authors concluded that although the results are encouraging, nonetheless, the correct application

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

of stem cell transplantation remains an area of active investigation; results of larger randomized, double-blind clinical trials, will certainly improve the knowledge of the appropriate clinical use of stem cell therapy in SSc patients.

Host et al (2017) stated that autologous hematopoietic stem cell transplantation (AHSCT) has emerged as a therapeutic option for patients with refractory, severe autoimmune disease. This is a systematic review of the current literature on AHSCT in adult patients with systemic sclerosis (SSc). Original articles published between 2005 and 2016 that evaluated the use of AHSCT in patients with SSc were reviewed with respect to the primary outcomes of overall and transplant related mortality (TRM) rates, and secondary outcomes of changes in mRSS, FVC, progression-free survival (PFS)/EF) and quality of life (QOL) measures. These researchers also focused on patient characteristics, the AHSCT conditioning and mobilization regimens used, and their relationship to patient outcome in each study. Of the 155 articles found, only 9 articles were suitable for review. There were 2 randomized controlled trials (RCTs), ASTIS and ASSIST, and 7 observational and cohort studies. In general, patients undergoing AHSCT had diffuse SSc with mRSS greater than 14, and interstitial lung disease (ILD). The 2 RCTs showed a benefit in PFS/EFS (80 to 81 %), FVC and QOL measures in AHSCT compared to monthly cyclophosphamide. All the studies showed an improvement in mRSS; TRM rates varied among studies, from 0 to 23 %, with a trend to higher mortality rates in studies using higher doses of cyclophosphamide or myeloablative conditioning regimens. The authors concluded that AHSCT is beneficial in some patients with SSc and that patient selection and conditioning regimens are critical determinants of prognosis and mortality post-ASCT.

Del papa et al (2017) retrospectively evaluated the efficacy of AHSCT in 18 patients with rapidly progressive diffuse cutaneous systemic sclerosis (rp-dcSSc), and compared their disease outcomes with those of 36 demographically- and clinically-matched patients treated with conventional therapies. Cutaneous involvement, by performing mRSS, lung diffusion capacity, by measuring diffusing capacity of lung for carbon monoxide (DLCO), and disease activity, by applying the European Scleroderma Study Group (ESSG) scoring system, were the outcome variables measured at the baseline time and then every 12 months for the following 60 months in both the AHSCT-treated patients and the control group. In the AHSCT group, TRM was 5.6 %. In this group, both mRSS and ESSG scores showed a significant reduction 1 year after AHSCT ($p < 0.002$); and these results were maintained until the end of follow-up. Conversely, DLCO values remained stable during the whole period of follow-up. Survival rate of AHSCT group was much higher than that observed in the whole control group ($p = 0.0005$). The probability that the ESSG score and mRSS would remain at a high level, and DLCO could decrease, was significantly higher in the control group as a whole and in the subgroup of control patients treated with cyclophosphamide than in the AHSCT group. The authors concluded that the findings of this study confirmed that the AHSCT was effective in prolonging survival, as well as in inducing a rapid reduction of skin involvement and disease activity, and preserving lung function in patients with rp-dcSSc.

The experts from the European League against Rheumatism (EULAR, 2017) recommended that hematopoietic stem cell transplantation (HSCT) should be considered for the treatment of selected patients with rapidly progressive SSc at risk of organ failure. To reduce the risk of treatment-related side effects, HSCT should be performed in selected centers with experience in this kind of treatment. Careful evaluation of the benefit to risk ratio in individual patients with SSc selected for HRCT should be done by experts. Further studies should help to identify subgroups of patients with SSc in whom HSCT would be most beneficial.

In a prospective, observational study, Nair and colleagues (2018) evaluated long-term outcomes in a cohort of patients with SSc treated with HSCT. A total of 4 SSc patients who underwent HSCT at a tertiary care center in India between 2008 to 2012 were included in this study. The selection criteria included young individuals with rapidly progressive disease and at least one major organ involvement. These researchers used granulocyte colony-stimulating factor for peripheral blood stem cell mobilization, pre-transplant conditioning with fludarabine, cyclophosphamide and rabbit anti-thymocyte globulin followed by re-infusion of autologous stem cells as per standard institute protocol. A total of 4 patients (1 male and 3 females) underwent autologous HSCT for SSc. Patients had heterogeneous disease manifestations including severe Raynaud's phenomenon with vasculopathic ulcers, gastro-intestinal (GI) problems and mild ILD. Patients were followed-up for a mean duration of 7 years. There was significant sustained improvement in skin score,

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

vasculopathy and GI manifestations; ILD did not show any deterioration. The QOL indices showed remarkable improvement in all subjects. No complications related to transplant were noted. The authors concluded that in the absence of an effective pharmacotherapy for SSc, autologous HSCT has a huge potential in management of cutaneous and internal organ manifestations.

Sullivan and colleagues (2018) compared myeloablative CD34+ selected autologous hematopoietic stem-cell transplantation with immunosuppression by means of 12 monthly infusions of cyclophosphamide in patients with scleroderma. These researchers randomly assigned adults (18 to 69 years of age; had scleroderma for 5 years or less with pulmonary or renal involvement) with severe scleroderma to undergo myeloablative autologous stem-cell transplantation (36 participants) or to receive cyclophosphamide (39 participants). The primary end-point was a global rank composite score comparing participants with each other on the basis of a hierarchy of disease features assessed at 54 months: death, event-free survival (EFS; survival without respiratory, renal, or cardiac failure), forced vital capacity (FVC), the score on the Disability Index of the Health Assessment Questionnaire, and the modified Rodnan skin score (mRSS). In the intention-to-treat population, global rank composite scores at 54 months showed the superiority of transplantation (67 % of 1,404 pair-wise comparisons favored transplantation and 33 % favored cyclophosphamide, p = 0.01). In the per-protocol population (participants who received a transplant or completed greater than or equal to 9 doses of cyclophosphamide), the rate of EFS at 54 months was 79 % in the transplantation group and 50 % in the cyclophosphamide group (p = 0.02). At 72 months, Kaplan-Meier estimates of EFS (74 % versus 47 %) and overall survival (OS; 86 % versus 51 %) also favored transplantation (p = 0.03 and 0.02, respectively). A total of 9 % of the participants in the transplantation group had initiated disease-modifying anti-rheumatic drugs (DMARDs) by 54 months, as compared with 44 % of those in the cyclophosphamide group (p = 0.001). Treatment-related mortality in the transplantation group was 3 % at 54 months and 6 % at 72 months, as compared with 0 % in the cyclophosphamide group. The authors concluded that myeloablative autologous hematopoietic stem-cell transplantation achieved long-term benefits in patients with scleroderma, including improved EFS and OS, at a cost of increased expected toxicity. Rates of treatment-related death and post-transplantation use of DMARDs were lower than those in previous reports of non-myeloablative transplantation.

Inclusion criteria of this trial included pulmonary involvement (required active interstitial lung disease (as determined by broncho-alveolar cell composition or ground-glass opacities on computed tomography of the chest) plus either a forced vital capacity (FVC) or a diffusing capacity of the lung for carbon monoxide (DLCO) of less than 70 % of the predicted value) as well as renal involvement required previous scleroderma-related renal disease. Key exclusion criteria included active gastric antral vascular ectasia, a DLCO of less than 40 % of the predicted value, an FVC of less than 45 % of the predicted value, a left ventricular ejection fraction (LVEF) of less than 50 %, a creatinine clearance of less than 40 ml/min, pulmonary arterial hypertension, or more than 6 months of previous treatment with cyclophosphamide.

Age-Related Macular Degeneration

Du et al (2011) stated that there is currently no Food and Drug Administration-approved therapy for treating patients with geographic atrophy, a late stage of age-related macular degeneration (AMD). Cell transplantation has the potential to restore vision in these patients. These researchers discussed how recent advancement in induced pluripotent stem (iPS) cells provides a promising therapy for GA treatment. Recent advances in stem cell biology have demonstrated that it is possible to derive iPS cells from human somatic cells by introducing re-programming factors. Human retinal pigment epithelium (RPE) cells and photoreceptors can be derived from iPS cells by defined factors. Studies show that transplanting these cells can stabilize or recover vision in animal models. However, cell derivation protocols and transplantation procedures still need to be optimized. Much validation has to be done before clinical-grade, patient-derived iPS can be applied for human therapy. For now, RPE cells and photoreceptors derived from patient-specific iPS cells can serve as a valuable tool in elucidating the mechanism of pathogenesis and drug discovery for AMD.

Huang et al (2011) noted that retinal degenerative diseases that target photoreceptors or the adjacent RPE affect millions of people worldwide. Retinal degeneration is found in many different forms of retinal diseases including retinitis pigmentosa, AMD, diabetic retinopathy, cataracts, and glaucoma. Effective

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

treatment for retinal degeneration has been widely investigated. Gene-replacement therapy has been shown to improve visual function in inherited retinal disease. However, this treatment was less effective with advanced disease. Stem cell-based therapy is being pursued as a potential alternative approach in the treatment of retinal degenerative diseases.

Schwartz et al (2012) stated that it has been 13 years since the discovery of human embryonic stem cells (hESCs). These researchers provided the first description of hESC-derived cells transplanted into human patients. They started 2 prospective clinical studies to establish the safety and tolerability of sub-retinal transplantation of hESC-derived retinal pigment epithelium (RPE) in patients with Stargardt's macular dystrophy and dry AMD -- the leading cause of blindness in the developed world. Pre-operative and post-operative ophthalmic examinations included visual acuity, fluorescein angiography, optical coherence tomography, and visual field testing. The authors concluded that hESC-derived RPE cells showed no signs of hyper-proliferation, tumorigenicity, ectopic tissue formation, or apparent rejection after 4 months. They stated that continued follow-up and future study is needed. The ultimate therapeutic goal will be to treat patients earlier in the disease processes, potentially increasing the likelihood of photoreceptor and central visual rescue.

Amyotrophic Lateral Sclerosis

Glass et al (2012) advances in stem cell biology have generated intense interest in the prospect of transplanting stem cells into the nervous system for the treatment of neurodegenerative diseases. These researchers reported the results of an ongoing phase I trial of intra-spinal injections of fetal-derived neural stems cells in patients with amyotrophic lateral sclerosis (ALS). This is a first in-human clinical trial with the goal of assessing the safety and tolerability of the surgical procedure, the introduction of stem cells into the spinal cord, and the use of immunosuppressant drugs in this patient population. A total of 12 patients received either 5 unilateral or 5 bilateral (10 total) injections into the lumbar spinal cord at a dose of 100,000 cells/injection. All patients tolerated the treatment without any long-term complications related to either the surgical procedure or the implantation of stem cells. Clinical assessments ranging from 6 to 18 months after transplantation demonstrated no evidence of acceleration of disease progression due to the intervention. One patient has shown improvement in his clinical status, though these data must be interpreted with caution since this trial was neither designed nor powered to measure treatment efficacy. These results allow the authors to report success in achieving the phase I goal of demonstrating safety of this therapeutic approach. Based on these positive results, the authors can now advance this trial by testing intra-spinal injections into the cervical spinal cord, with the goal of protecting motor neuron pools affecting respiratory function, which may prolong life for patients with ALS.

Martinez et al (2012) stated that ALS is characterized by the selective death of motor neurons. Stem cells have been proposed as a potential therapeutic strategy. These researchers described the safety of stem cell transplantation into the frontal motor cortex to improve upper motor neuron function. A total of 67 patients with definite ALS were included. After giving their informed consent, the patients underwent magnetic resonance imaging, functional rating, pulmonary function test, and laboratory tests. Their bone marrow was stimulated with daily filgrastim (300 µg) given subcutaneously for 3 days. Peripheral blood mononuclear cells were obtained by leukapheresis. Isolated CD133+ stem cells were suspended in 300 µl of the patient's cerebrospinal fluid and implanted into the motor cortex. Adverse events were recorded at each step of the procedure and were classified according to the Common Terminology Criteria for Adverse Events v3.0. The survival at 1 year was 90 % after transplantation with a mean long-term survival rate of 40.17 months from diagnosis. The most common adverse events were in grades I to II and involved transient skin pain (19.5 % of patients) attributed to the insertion of the Mahurkar catheter into the subclavian vein, minor scalp pain (15.9 %), and headache (12.2 %) from the surgical procedure. Several patients (1.5 % to 4.5%) reported diverse grade I adverse events. There were 2 deaths, 1 considered to be associated with the procedure (1.5 %) and the other associated with the disease. Autologous stem cell transplantation into the frontal motor cortex is safe and well-tolerated by patients. Moreover, they stated that further controlled studies are needed to define the efficacy of this procedure.

Thomsen et al (2014) noted that while the genetics of ALS are becoming more understood in familial cases, the mechanisms underlying disease pathology remain unclear and there are no effective treatment options.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Without understanding what causes ALS it is difficult to design treatments. However, in recent years stem cell transplantation has emerged as a potential new therapy for ALS patients. While motor neuron replacement remains a focus of some studies trying to treat ALS with stem cells, there is more rationale for using stem cells as support cells for dying motor neurons as they are already connected to the muscle. This could be through reducing inflammation, releasing growth factors, and other potential less understood mechanisms. Prior to moving into patients, stringent pre-clinical studies are required that have at least some rationale and effectiveness in animal models and good safety profiles. However, given the poor understanding of what causes ALS and whether stem cells may ameliorate symptoms, there should be a push to determine cell safety in pre-clinical models and then a quick translation to the clinic where patient trials will show if there is any effectiveness. The authors provided a critical review of current clinical trials using either mesenchymal stromal cells (MSCs) or neural stem cells to treat ALS patients. Pre-clinical data leading to these trials, as well as those in development were also evaluated in terms of mechanisms of action, validity of conclusions and rationale for advancing stem cell treatment strategies for ALS.

In an open-label, phase II study, Glass and colleagues (2016) tested the safety of spinal cord transplantation of human stem cells in patients with ALS with escalating doses and expansion of the trial to multiple clinical centers. This trial included 15 participants at 3 academic centers divided into 5 treatment groups receiving increasing doses of stem cells by increasing numbers of cells/injection and increasing numbers of injections. All subjects received bilateral injections into the cervical spinal cord (C3 to C5). The final group received injections into both the lumbar (L2 to L4) and cervical cord through 2 separate surgical procedures. Subjects were evaluated for adverse events (AEs) and progression of disease, as measured by the ALS Functional Rating Scale-Revised, forced vital capacity, and quantitative measures of strength. Statistical analysis focused on the slopes of decline of these phase II trial participants alone or in combination with the phase I participants (previously reported), comparing these groups to 3 separate historical control groups. Adverse events were mostly related to transient pain associated with surgery and to side effects of immunosuppressant medications. There was 1 incident of acute post-operative deterioration in neurologic function and another incident of a central pain syndrome. These researchers could not discern differences in surgical outcomes between surgeons. Comparisons of the slopes of decline with the 3 separate historical control groups showed no differences in mean rates of progression. The authors concluded that intra-spinal transplantation of human spinal cord-derived neural stem cells can be safely accomplished at high doses, including successive lumbar and cervical procedures. The procedure can be expanded safely to multiple surgical centers. This study provided Class IV evidence that for patients with ALS, spinal cord transplantation of human stem cells can be safely accomplished and did not accelerate the progression of the disease. This study lacked the precision to exclude important benefit or safety issues.

In an editorial that accompanied the aforementioned study, Appel and Armon (2016) stated that these are clearly early stages of evaluating the risks and benefits of transplanting neural progenitor stem cells in patients with ALS.

Chronic Inflammatory Demyelinating Polyradiculopathy

Snowden et al (2012) provided revised and updated guidelines of the European Group for Blood and Marrow Transplantation (EBMT) for both the current application and future development of HSCT in ADs in relation to the benefits, risks and health economic considerations of other modern treatments. These investigators listed sibling donor and well-matched unrelated donor as generally not recommended for chronic inflammatory demyelinating polyradiculopathy and neuromyelitis optica, among many ADs.
They emphasized a need for prospective interventional and non-interventional studies, where feasible, along with systematic data reporting, in accordance with EBMT policies and procedures.

Press et al (2014) noted that only 70 to 80 % of patients with chronic inflammatory demyelinating polyneuropathy (CIDP) respond satisfactorily to the established first-line immunomodulatory treatments. Autologous HSCT (AHSCT) has been performed as a last treatment resort in a few therapy-refractory cases with CIDP. These researchers described the results of AHSCT in 11 consecutive Swedish patients with therapy-refractory CIDP with a median follow-up time of 28 months. Case data were gathered retrospectively for AHSCT treatments in 11 patients with CIDP refractory to the first-line immunomodulatory treatments, intravenous high-dose immunoglobulin, corticosteroids and plasma exchange

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

and to 1 or more second-line treatments used in 10 of the 11 patients. The median Inflammatory Neuropathy Cause and Treatment (INCAT) score within 1 month prior to AHSCT was 6 and the Rankin score 4. Total INCAT and Rankin scores improved significantly within 2 to 6 months after AHSCT and continued to do so at last follow-up. The motor action potential amplitudes (CMAP) improved already within 4 months (median) after AHSCT; 3 of the 11 patients relapsed during the follow-up period, requiring re-transplantation with AHSCT in 1; 8 of the 11 patients maintained drug-free remission upon last follow-up. Autologous HSCT was safe but on the short-term associated with a risk of cytomegalovirus (CMV) and Epstein-Barr virus re-activation, CMV disease, hemorrhagic cystitis and pancreatitis. The authors concluded that these findings though hampered by the limited number of patients and the lack of a control group suggested AHSCT to be effective in therapy-refractory CIDP, with a manageable complication profile. They stated that confirmation of these results is needed through randomized controlled trials (RCTs).

Zheng et al (2023) noted that therapeutic options for CIDP include IVIG, plasmapheresis (PE), corticosteroids and immunosuppressive drugs; however, many patients with CIDP remain refractory to treatment and develop severe functional disability. In a systematic review and a meta-analysis, these investigators examined the effectiveness of HSCT in refractory CIDP patients. The study was based on queries in the PubMed, Cochrane Central Register of Controlled Trials, Embase, Web of Science and clinicaltrials.gov databases on December 4, 2022. Studies that met the eligibility criteria were included after screening. Patients' characteristics, treatment regimens, and outcome measures were extracted. A total of 89 patients in 11 studies were included. The pooled estimate of responsiveness among the 4 included studies was 87.04 % (95 % CI: 66.7 % to 99.5 %) and the pooled estimate of freedom of all immune modulating or suppressive drugs was 80.75 % (95 % CI: 71.2 % to 90.2 %). The authors concluded that this meta-analysis and systematic review suggested that HSCT could be effective in the treatment of refractory CIDP. W Moreover, these researchers stated that while there are risks involved, HSCT may be a beneficial and viable therapy for refractory CIDP when carefully evaluated.

Crohn's Disease

Leung and associates (2006) reviewed the evidence regarding the use of hematopoietic cell transplantation (HCT) for the treatment of Crohn's disease (CD). A Medline search (1970 to 2005) was performed using the keywords -- bone marrow transplant, stem cell, hematopoietic cell, Crohn's disease and inflammatory bowel disease. These researchers identified 1 case in which a patient developed CD following an allogeneic HCT from a sibling suffering with CD. Evidence for transfer of the genetic predisposition to develop CD was also identified with report of a patient that developed severe CD following an allogeneic HCT. Following HCT it was found that the donor (that had no signs or symptoms of CD) and the recipient had several haplotype mismatches in HLA class III genes in the IBD3 locus including a polymorphism of NOD2/CARD15 that has been associated with CD. A total of 33 published cases of patients with CD who underwent either autologous or allogeneic HCT were identified. At the time of publication 29 of these 33 patients were considered to be in remission. The median follow-up time was 7 years, and 20 months for allogeneic and autologous HCT, respectively. For patients who underwent HCT primarily for treatment of their CD there have been no mortalities related to transplant complications. The authors concluded that these preliminary data suggested that both allogeneic and autologous HCT may be effective in inducing remission in refractory CD. This supports the hypothesis that the hematolymphatic cells play a key role in CD and that re-setting of the immune system may be a critical approach in the management or cure of CD.

Al-toma et al (2007) reported on the feasibility, safety, and effectiveness of ASCT in patients with refractory celiac disease with aberrant T cells (RCD type II). A total of 13 patients with RCD type II were evaluated. Seven patients (3 women, 4 men, mean age of 61.5 years [range of 51 to 69 years]) underwent transplantation. After conditioning with fludarabine and melphalan, ASCT was performed. Patients were monitored for response, adverse effects, and hematopoietic reconstitution. All 7 patients completed the mobilization and leukapheresis procedures successfully and subsequently underwent conditioning and transplantation. Engraftment occurred in all patients. No major non-hematological toxicity or transplantation-related mortality was observed. There was a significant reduction in the aberrant T cells in duodenal biopsies associated with improvement in clinical well-being and normalization of hematological and biochemical markers (mean follow-up of 15.5 months; range of 7 to 30 months). One patient died 8

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

months after transplantation from progressive neuro-celiac disease. The authors concluded that these preliminary results showed that HDC followed by ASCT seems feasible and safe and might result in long-term improvement of patients with RCD type II whose condition did not respond promptly to available drugs.

Kotlarz et al (2012) examined heterogeneity among patients with very early onset inflammatory bowel disease (IBD), its mechanisms, and the use of allogeneic hematopoietic stem cell transplantation (HSCT) to treat this disorder. These investigators analyzed 66 patients with early onset IBD (younger than 5 years of age) for mutations in the genes encoding IL-10, IL-10R1, and IL-10R2. IL-10R deficiency was confirmed by functional assays on patients' peripheral blood mononuclear cells (immunoblot and enzyme-linked immunosorbent assay analyses). They assessed the therapeutic effects of standardized allogeneic HSCT. Using a candidate gene sequencing approach, these researchers identified 16 patients with IL-10 or IL-10R deficiency: 3 patients had mutations in IL-10, 5 had mutations in IL-10R1, and 8 had mutations in IL-10R2. Refractory colitis became manifest in all patients within the first 3 months of life and was associated with perianal disease (16 of 16 patients). Extra-intestinal symptoms included folliculitis (11 of 16) and arthritis (4 of 16). Allogeneic HSCT was performed in 5 patients and induced sustained clinical remission with a median follow-up time of 2 years. In-vitro experiments confirmed reconstitution of IL-10R-mediated signaling in all patients who received the transplant. The authors concluded that they identified loss of function mutations in IL-10 and IL-10R in patients with very early onset IBD. These findings indicated that infantile IBD patients with perianal disease should be screened for IL-10 and IL-10R deficiency and that allogeneic HSCT can induce remission in those with IL-10R deficiency.

In an editorial that accompanied the study by Koltarz et al, Muise and colleagues (2012) explained that the short-term results of this small series should be interpreted with caution. The editorialists noted that “This study also reports the outcomes of 5 patients after hematopoietic stem cell transplantation (HSCT) with 2-year follow-up on 4 patients. The authors used a highly immunosuppressive conditioning regimen that also resulted in depletion of myeloid cells including dendritic cells in conjunction with intense gut decolonization. HSCT in this patient population was remarkably well tolerated and led to complete clinical remission in 4 out of 5 patients. Although successful, it will be interesting to determine whether HSCT will result in a permanent “cure” for the colonic, joint, and skin disease in patients with IL-10RB mutations, because this receptor is widely expressed in these tissues and presumably the deficiency would persist after HSCT. Variable results in HSCT ability to establish long-term remission in adult-onset IBD patients (reviewed by Anderson et al) points to cautious interpretation of these promising short-term HSCT results. This study also leads to another important question regarding the role of HSCT in all infantile IBD. As the authors correctly point out, there is a strong possibility that some infant and VEO-IBD patients may have gene defects in the non-hematopoietic cells, including epithelial barrier genes and, therefore, HSCT for all infantile IBD cannot be recommended at this time, unless a hematopoietic gene mutation that is functionally linked to disease can be identified”.

The National Institute for Health and Clinical Excellence’s clinical guideline on “Crohn's disease: Management in adults, children and young people” (NICE, 2012) does not mention the use of stem cell transplantation as a therapeutic option. Furthermore, a RCT on the use of autologous stem cell transplantation for CD is currently underway.

Al-toma et al (2014) stated that both, autologous and allogeneic HSCT can be used to cure or ameliorate a variety of malignant and non-malignant diseases. The rationale behind this strategy is based on the concept of immuno-ablation using HDC, with subsequent regeneration of naive T-lymphocytes derived from re-infused hematopoietic progenitor cells. In addition, the use of HSCT allows for the administration of HDC (whether or not combined with immunomodulating agents such as anti-thymocyte globulin) resulting in a prompt remission in therapy-refractory patients. These researchers provided an update of the major areas of successful uses of HSCT in non-malignant gastro-intestinal disorders. A Medline search has been conducted and all relevant published data were analyzed. Hematopoietic stem cell transplantation has been proved successful in treating refractory CD. Patients with refractory celiac disease type II and a high risk of developing enteropathy associated T-cell lymphoma have shown promising improvement. Data concerning HSCT and mesenchymal SCT in end-stage chronic liver diseases are encouraging. In refractory autoimmune gastro-intestinal diseases HDC followed by HSCT seems feasible and safe and might result in long-term

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

improvement of disease activity. Moreover, they stated that mesenchymal SCT for a selected group of CD is promising and may represent a significant therapeutic alternative in treating fistulas in CD.

Hawkey et al (2015) evaluated the effect of autologous HSCT on refractory CD. This was a parallel-group randomized clinical trial conducted in 11 European transplant units from July 2007 to September 2011, with follow-up through March 2013. Patients were aged 18 to 50 years with impaired quality of life from refractory CD not amenable to surgery despite treatment with 3 or more immunosuppressive or biologic agents and corticosteroids. All patients underwent stem cell mobilization before 1:1 randomization to immune-ablation and HSCT (n = 23) or control treatment (HSCT deferred for 1 year [n = 22]). All were given standard CD treatment as needed. Primary outcomes included sustained disease remission at 1 year, a composite primary end-point comprising clinical remission (Crohn Disease Activity Index (CDAI) less than 150 [range of 0 to 600]), no use of corticosteroids or immunosuppressive or biologic drugs for at least the last 3 months, and no endoscopic or radiological evidence of active (erosive) disease anywhere in the gastro-intestinal (GI) tract. Secondary outcomes were individual components of the primary composite outcome and other measures of disease activity, laboratory results, quality of life and functional status, and GI tract imaging. A total of 23 patients underwent HSCT and 22 received standard CD treatment (controls). Sustained disease remission was achieved in 2 patients undergoing HSCT (8.7 %) versus 1 control patient (4.5 %) (absolute difference, 4.2 % [95 % confidence interval [CI]: -14.2 % to 22.6 %]; p = 0.60); 14 patients undergoing HSCT (61 %) versus 5 control patients (23 %) had discontinued immunosuppressive or biologic agents or corticosteroids for at least 3 months (difference, 38.1 % [95 % CI: 9.3 % to 59.3 %]; p = 0.01); 10 versus 2 patients had a CDAI less than 150 (remission) at the final evaluation, 8 (34.8 %) versus 2 (9.1 %) for 3 or more months (difference, 25.7 % [95 % CI: 1.1 % to 47.1 %]; p = 0.052); 8 (34.8 %) versus 2 (9.1 %) patients were adjudicated free of active disease on endoscopy and radiology at final assessment (difference, 25.7 % [95 % CI: 1.1 % to 47.1 %]; p = 0.054). There were 76 serious adverse events in patients undergoing HSCT versus 38 in controls; 1 patient undergoing HSCT died. The authors concluded that among adult patients with refractory CD not amenable to surgery who had impaired quality of life, HSCT, compared with conventional therapy, did not result in a statistically significant improvement in sustained disease remission at 1 year and was associated with significant toxicity. These findings do not support the widespread use of HSCT for patients with refractory CD.

In a multi-center retrospective analysis, Brierley and colleagues (2018) examined the safety and safety for patients undergoing AHSCT for CD in Europe outside the ASTIC trial. These researchers identified 99 patients in the European Society for Blood and Marrow Transplantation (EBMT) registry who were eligible for inclusion. Transplant and clinical outcomes were obtained for 82 patients from 19 centers in 7 countries. Median patient age was 30 years (range of 20 to 65). Patients had failed or been intolerant to a median of 6 lines of drug therapy; 61/82 (74 %) had had surgery. Following AHSCT, 53/78 (68 %) experienced complete remission or significant improvement in symptoms at a median follow-up of 41 months (range of 6 to 174); 22/82 (27 %) required no medical therapy at any point post-AHSCT. In patients who had re-started medical therapy at last follow-up, 57 % (24/42) achieved remission or significant symptomatic improvement with therapies to which they had previously lost response or been non-responsive. Treatment-free survival at 1 year was 54 %. On multi-variate analysis, peri-anal disease was associated with adverse treatment-free survival (HR 2.34, 95 % CI: 1.14 to 4.83, p = 0.02); 1 patient died due to infectious complications (CMV disease) at day +56. The authors concluded that AHSCT was relatively safe and appeared to be effective in controlling otherwise treatment-resistant CD. These researchers stated that further prospective RCTs against standard of care (SOC) are needed.

In a retrospective study, Hernanz and associates (2019) described findings on patients (n = 7) with refractory CD subjected to AHSCT; 3 patients (43 %) presented with clinical and endoscopic remission; 1 patient (14 %) clinical improvement without remission, and 3 patients (43 %) remained active with the need to re-start treatment in the assessment of the initial response to the AHSCT (after 6 months). Symptoms recurred in 5 of the 7 patients (71 %) and all of them had to re-start medical treatment after an average of 13.8 months (range of 3 to 30 months). Only 1 patient needed surgery after the AHSCT. At the end of the follow-up, after a mean of 48 months (range of 17 to 78 months), 5/7 (71 %) of the patients were in clinical remission with or without treatment. The authors concluded that AHSCT may be a promising therapeutic option for patients with refractory CD.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Diabetes Mellitus

In a prospective phase I/II study, Voltarelli and co-workers (2007) examined the safety and metabolic effects of high-dose immunosuppression followed by autologous non-myeloablative hematopoietic stem cell transplantation (AHST) in newly diagnosed type 1 diabetes mellitus (T1DM). A total of 15 patients with type 1 DM (aged 14 to 31 years) diagnosed within the previous 6 weeks by clinical findings and hyperglycemia and confirmed with positive antibodies against glutamic acid decarboxylase were included. Patients with previous diabetic ketoacidosis were excluded after the first patient with diabetic ketoacidosis failed to benefit from AHST. Hematopoietic stem cells were mobilized with cyclophosphamide (2.0 g/m2) and granulocyte colony-stimulating factor (10 microg/kg per day) and then collected from peripheral blood by leukapheresis and cryopreserved. The cells were injected intravenously after conditioning with cyclophosphamide (200 mg/kg) and rabbit anti-thymocyte globulin (4.5 mg/kg). Main outcome measures were morbidity and mortality from transplantation and temporal changes in exogenous insulin requirements (daily dose and duration of usage). Secondary end points included serum levels of hemoglobin A1c, C-peptide levels during the mixed-meal tolerance test, and anti-glutamic acid decarboxylase antibody titers measured before and at different times following AHST. During a 7- to 36-month follow-up (mean of 18.8 months), 14 patients became insulin-free (1 for 35 months, 4 for at least 21 months, 7 for at least 6 months; and 2 with late response were insulin-free for 1 and 5 months, respectively). Among those, 1 patient resumed insulin use 1 year after AHST. At 6 months after AHST, mean total area under the C-peptide response curve was significantly greater than the pre-treatment values, and at 12 and 24 months it did not change. Anti-glutamic acid decarboxylase antibody levels decreased after 6 months and stabilized at 12 and 24 months. Serum levels of hemoglobin A(1c) were maintained at less than 7 % in 13 of 14 patients. The only acute severe adverse effect was culture-negative bilateral pneumonia in 1 patient and late endocrine dysfunction (hypothyroidism or hypogonadism) in 2 others. There was no mortality. The authors noted that high-dose immunosuppression and AHST were performed with acceptable toxicity in a small number of patients with newly diagnosed type 1 DM. With AHST, beta cell function was increased in all but 1 patient and induced prolonged insulin independence in the majority of the patients. They stated that further follow-up is needed to confirm the duration of of insulin independence and the mechanisms of action of the procedure. Furthermore, randomized controlled studies and more biological studies are needed to confirm the role of this procedure in changing the natural history of type 1 DM and to assess the contribution of hematopoietic stem cells to this change.

El-Badawy and El-Badri (2016) evaluated the clinical evidence on the safety and effectiveness of different types of stem cell therapy for both T1DM and T2DM. These investigators pooled participant-level data from 22 eligible clinical trials that satisfied inclusion criteria, with a total of 524 patients. There were significant differences in the outcome based on the type and source of the infused cells. Out of all T1DM patients who received CD34+ hematopoietic stem cell (HSC) infusion, 58.9 % became insulin-independent for a mean period of 16 months, whereas the results were uniformly negative in patients who received umbilical cord blood (UCB). Infusion of umbilical cord MSCs (UC-MSCs) provided significantly beneficial outcome in T1DM, when compared to bone-marrow mesenchymal stem cells (BM-MSCs) ($p < 0.0001$ and $p = 0.1557$). Administration of stem cell therapy early after DM diagnosis was more effective than intervention at later stages (relative risk [RR] = 2.0, $p = 0.0008$). Adverse events (AEs) were observed in only 21.72 % of both T1DM and T2DM stem cell recipients with no reported mortality. Out of all poor responders, 79.5 % were diagnosed with diabetic ketoacidosis. The authors concluded that:

I. remission of DM is possible following stem cell therapy;
II. stem cell transplantation can be a safe and effective approach for therapy of DM;
III. available data from these clinical trials indicate that the most promising therapeutic outcome was shown in mobilized marrow CD34+ HSCs;
IV. patients with previously diagnosed diabetic ketoacidosis are not good candidates for the applied approaches stem cell therapy;
V. stem cell therapy at early stages after DM diagnosis is more effective than intervention at later stages; and
VI. well-designed large scale randomized studies considering the stem cell type, cell number, and infusion method in DM patients are urgently needed.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

I. remission of DM is possible following stem cell therapy;
II. stem cell transplantation can be a safe and effective approach for therapy of DM;
III. available data from these clinical trials indicate that the most promising therapeutic outcome was shown in mobilized marrow CD34+ HSCs;
IV. patients with previously diagnosed diabetic ketoacidosis are not good candidates for the applied approaches stem cell therapy;
V. stem cell therapy at early stages after DM diagnosis is more effective than intervention at later stages; and
VI. well-designed large scale randomized studies considering the stem cell type, cell number, and infusion method in DM patients are urgently needed.

This meta-analysis had several drawbacks:

I. some of the cases covered in this meta-analysis have been evaluated in a single study with low statistical power. In some studies, low number of patients enrolled, not sufficient for significant interpretation,
II. the total sample size was not large and the follow-up time was not adequately long, and
III. the number of trials was relatively small, and more accurate recommendations could be made when analyzing a larger cohort.

I. some of the cases covered in this meta-analysis have been evaluated in a single study with low statistical power. In some studies, low number of patients enrolled, not sufficient for significant interpretation,
II. the total sample size was not large and the follow-up time was not adequately long, and
III. the number of trials was relatively small, and more accurate recommendations could be made when analyzing a larger cohort.

Altogether, these limitations reflected the paucity of the available published data on this very important and much anticipated form of cell therapy and indicated the large gap between the pre-clinical and clinical studies.

He et al (2021) stated that although many studies have examined the treatment of diabetes with stem cells, no articles have been published to summarize the safety and effectiveness of MSCs in RCTs. In a systematic review, these investigators examined the evidence from RCTs and, where possible, carried out meta-analyses to provide a reliable numerical summary and the most comprehensive assessment of safety and effectiveness with MSCs in diabetes. PubMed, Web of Science, Ovid, the Cochrane Library and CNKI were searched. The retrieval time was from establishment of these databases to January 4, 2020. A total of 7 RCTs were eligible for analysis, including 413 participants. Meta-analysis results showed that there were no significant differences in the reduction of fasting plasma glucose (FPG) compared to the baseline [mean difference (MD) = -1.05, 95 % CI: -2.26 to 0.16), $p < 0.01$, I2 = 94 %] and the control group [MD = -0.62, 95 % CI: -1.46 to 0.23), $p < 0.01$, I2 = 87 %]. The MSCs treatment group showed a significant decrease in hemoglobin (Hb) A1c [random-effects, MD = -1.32, 95 % CI: -2.06 to -0.57, $p < 0.01$, I2 = 90 %] after treatment. Furthermore, HbA1c reduced more significantly in MSC treatment group than in control group [random-effects, MD = -0.87, 95 % CI : -1.53 to -0.22, $p < 0.01$, I2 = 82 %] at the end of follow-up. However, as for fasting C-peptide levels, the estimated pooled MD showed that there was no significant increase [MD = -0.07, 95 % CI: -0.30 to 0.16, $p < 0.01$, I2 = 94 %] in MSCs treatment group compared with that in control group. Notably, there was no significant difference in the incidence of AEs between MSCs treatment group and control group (RR = 0.98, 95 % CI: 0.72 to 1.32, p = 0.02, I2 = 70 %). The most commonly observed adverse reaction in the MSC treatment group was hypoglycemia (29.95 %). The authors concluded that the findings of this meta-analysis revealed MSCs therapy may be a safe and effective intervention in subjects with diabetes; however, due to the limited studies, a number of high-quality as well as large-scale RCTs should be performed to confirm these conclusions.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Madani et al (2022) noted that stem cell transplantation (SCT) has been examined in the treatment of T1DM; however, previous studies have not presented a comprehensive view of its effect on treatment of T1DM. After registration of the present systematic review and meta-analysis in the PROSPERO, a search was carried out according to the Cochrane guidelines for evaluation of clinical trials to find eligible clinical trials that examined the effect of SCT on T1DM (based on ADA diagnostic criteria) from PubMed, Web of science, Scopus, etc., as well as registries of clinical trials from January 1, 2000, to September 31, 2019. A search strategy was designed using MeSH and EM-tree terms. Primary outcome included the changes in the insulin total daily dose (TDD) (U/kg) level, and secondary outcomes included the changes in the HbA1c, c-peptide, and adjusted HbA1c levels. The Q Cochrane test and I2 statistic were performed to assess the heterogeneity and its severity in primary clinical trials. The Cochrane ROB was used to determine risk of bias, and Cochrane Handbook for Systematic Reviews of Interventions was used in the full text papers. The meta-analysis was accomplished in the STATA software, and the results were shown on their forest plots. Confounders were evaluated by the meta-regression test. A total of 9,452 studies were electronically screened, and 35 papers were included for data extraction. The results of this review study showed that 173 (26.5 %) diabetic patients experienced insulin-free period (from 1 to 80 months), and 445 (68 %) showed reduction in TDD of insulin after the SCT. Combination of hematopoietic stem cell (HSC) with mesenchymal stem cell (MSC) transplantation were significantly associated with improvement of the TDD (SMD: - 0.586, 95 % CI: - 1.204 to - 0.509, I2: 0 %), HbA1c (SMD: - 0.736, 95 % CI: - 1.107 to - 0.365, I2: 0 %), adjusted HbA1c (SMD: - 2.041, 95 % CI: - 2.648 to - 1.434, I2: 38.4 %), and c-peptide (SMD: 1.917, 95 % CI: 0.192 to 3.641, I2: 92.5 %) on month 3 of follow-up, while its association had a growing trend from 3 to 12 months after the transplantation. Considering severe AEs, HSC transplantation accompanied with conditioning could not be suggested as a safe treatment. The authors concluded that most of the clinical trials of SCT in T1DM were single-arm studies. Although meta-analysis illustrated the SCT was associated with T1DM improvement, well-designed randomized clinical trials are needed to clarify its effectiveness. These researchers stated that the lack of well-designed clinical trials, control group, randomization, and blinding in almost all the studies on this topic were among the most important drawbacks.

In a systematic review, Liu et al (2024) examined the effectiveness of MSCs in the treatment of T1DM. Studies regarding the effects of MSCs for T1DM were retrieved in PubMed, Web of Science, Embase, and the Cochrane Library databases up to July 2023. Additional relevant studies were manually searched through citations. HbA1c, FBG, PBG, insulin requirement and C-peptide were assessed. The risk of bias was evaluated with the ROB 2.0 and ROBINS-I tools. A total of 6 RCTs and 8 non-RCTs were included. Of the 14 studies included, 2 examined BM-MSCs, 3 evaluated UC-MSCs, 5 evaluated AHSCT, 2 evaluated cord blood-derived stem cells (CB-SCs), and 2 evaluated umbilical cord-derived stem cells (UC-SCs) plus autologous bone marrow-derived mononuclear cells (aBM-MNCs). At the end of follow-up, a total of 10 studies found that MSCs improved glycemic outcomes in T1DM, while the remaining 4 studies showed no significant improvement. These researchers stated that findings supported the positive impacts observed from using MSCs in patients with T1DM. However, the overall methodological quality of the identified studies and findings was heterogeneous, limiting the interpretation of the therapeutic benefits of MSCs in T1DM. They stated that methodically rigorous research is needed to further increase credibility. The authors concluded that this systematic review highlighted the positive impacts observed from employing MSCs in patients with T1DM. The potential therapeutic value of MSCs remains promising; however, further investigation is needed due to variations in sample sizes and concerns regarding study quality. To get a clearer picture of the underlying mechanisms and establish whether MSCs are really beneficial to patients with T1DM, additional experimental and clinical studies with comprehensive baseline assessments and extended follow-up are needed.

Multiple Sclerosis

A recent study (Mancardi et al, 2001) on autologous hematopoietic stem cell (ASCT) transplantation for the treatment of patients with rapidly evolving secondary progressive MS concluded that the final impact of this procedure on disease course remains to be established. In a study (Verburg et al, 2001) to evaluate the effectiveness of HDC followed by ASCT in the treatment of patients with refractory, progressively erosive RA, the authors concluded that there is a need for randomized clinical trials (RCTs). Tyndall (2001) stated that randomized, prospective controlled phase III trials are needed to ascertain the effectiveness of ASCT

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

following immunoablation for the treatment of SLE, however, more phase I and II data is needed to plan the optimal protocol. An assessment by the BlueCross BlueShield Association Technology Evaluation Center (2001) concluded that there is insufficient evidence on health outcomes of the effect of stem cell transplant in autoimmune disease.

In a phase I/II study, Burt and colleagues (2009) examined the effects of autologous non-myeloablative hemopoietic stem cell transplantation in relapsing-remitting MS. Eligible patients had relapsing-remitting MS, attended Northwestern Memorial Hospital, and despite treatment with interferon beta had had 2 corticosteroid-treated relapses within the previous 12 months, or 1 relapse and gadolinium-enhancing lesions seen on magnetic resonance imaging (MRI) and separate from the relapse. Peripheral blood hemopoietic stem cells were mobilized with 2 g/m2 cyclophosphamide and 10 microg/kg per day filgrastim. The conditioning regimen for the hemopoietic stem cells was 200 mg/kg cyclophosphamide and either 20 mg alemtuzumab or 6 mg/kg rabbit anti-thymocyte globulin. Primary outcomes were progression-free survival and reversal of neurological disability at 3 years post-transplantation. These researchers also investigated the safety and tolerability of autologous non-myeloablative hemopoietic stem cell transplantation. A total of 21 patients were treated. Engraftment of white blood cells and platelets was on median day 9 (range of day 8 to 11) and patients were discharged from hospital on mean day 11 (range of day 8 to 13). One patient had diarrhea due to clostridium difficile and 2 patients had dermatomal zoster. Two of the 17 patients receiving alemtuzumab developed late immune thrombocytopenic purpura that remitted with standard therapy. A total of 17 of 21 patients (81 %) improved by at least 1 point on the Kurtzke expanded disability status scale (EDSS), and 5 patients (24 %) relapsed but achieved remission after further immunosuppression. After a mean of 37 months (range of 24 to 48 months), all patients were free from progression (no deterioration in EDSS score), and 16 were free of relapses. Significant improvements were noted in neurological disability, as determined by EDSS score ($p < 0.0001$), neurological rating scale score ($p = 0.0001$), paced auditory serial addition test ($p = 0.014$), 25-foot walk ($p < 0.0001$), and quality of life, as measured with the short form-36 questionnaire ($p < 0.0001$). The authors concluded that non-myeloablative autologous hemopoietic stem cell transplantation in patients with relapsing-remitting MS reverses neurological deficits, but these results need to be confirmed in a randomized trial.

Tappenden and colleagues (2010) stated that therapeutic options for secondary progressive MS (SPMS) are limited. Mitoxantrone is routinely used to stabilize disease progression; however, evolving evidence suggests clinical benefit from intensive treatment with autologous hematopoietic stem cell transplantation (HSCT). Given differences in cost and outcomes, preliminary cost-effectiveness studies are warranted if this approach is to be developed for more widespread application in SPMS. These researchers developed a decision-analytic Markov model to explore the potential cost-effectiveness of autologous HSCT versus mitoxantrone in SPMS, using patient-level data from registry sources. The model evaluates the lifetime costs and health outcomes associated with disability progression and relapse. Sensitivity analyses were undertaken to examine the uncertainty surrounding cost-effectiveness outcomes. In the absence of RCT evidence, conditions for comparative analysis were not ideal. Under optimistic assumptions, HSCT is estimated to cost below 3,000 pounds per quality adjusted life year gained. However, when a strict 6-month sustained progression rule is adopted, HSCT may be less effective and more expensive than mitoxantrone. The model results were sensitive to reducing procedural costs and HSCT-related mortality. The authors concluded that HSCT could potentially achieve an acceptable level of cost-effectiveness. However, caution should be exercised as large, high-quality RCTs comparing HSCT versus mitoxantrone are needed to validate these findings. Further analyses are necessary to examine the economic value of HSCT for the treatment of rapidly progressing, relapsing-remitting and aggressive forms of MS.

Atkin (2010) noted that MS is the leading autoimmune indication for autologous HSCT (aHSCT). Patient selection criteria and transplant interventions have been refined through a series of cohort and registry studies. High- and low-intensity chemotherapy-based conditioning regimens have been used, creating trade-offs between toxicity and effectiveness. Total body irradiation has been associated with greater toxicity and poor outcomes. Autologous HSCT stops MS relapses and lessens the disability in malignant MS, which otherwise rapidly incapacitates patients. Better responses occur in progressive MS earlier in the disease when it has a more inflammatory nature. Autologous HSCT prevents further disability in many patients, but some actually recover from their infirmities. Current regimens and supportive care result in very low morbidity and mortality. Patients with MS experience unique complications in addition to the expected

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

toxicities. Cytokines used alone for stem-cell mobilization may induce MS flares but are safe to be used in combination with steroids or cytotoxic agents. Urinary tract infections, herpes virus reactivation and an engraftment syndrome may occur early after aHSCT. Rarely secondary autoimmune diseases have been reported late after HSCT. Increasing experience in caring for patients with MS has reduced the frequency and severity of toxicity. Conceived as an opportunity to "reboot" a tolerant immune system, aHSCT is successful in treating patients with MS that is refractory to conventional immunomodulatory drugs. It is noted that patients are unlikely to benefit from aHSCT if they have had longstanding MS or disabilities that have advanced to the point where they are wheelchair bound. Furthermore, previous MS treatments can potentially impact aHSCT (e.g., interferon may impair stem cell collection, natalizumab may increase the risk of opportunistic infections, and mitoxantrone could cause myelodysplasia). Moreover, the author stated that aHSCT for MS is still controversial. It has not entered widespread clinical use because of its perceived high mortality, high cost, and competition from new pharmaceutical agents. The author noted that a direct comparison of aHSCT with conventional treatments in multi-center randomized controlled trials and publication of 5 to 10 years long-term follow-up data from cohort studies and international registries are a necessary step in further defining the role of aHSCT in the management of MS.

At present, there are no well-designed randomized controlled clinical trials of stem cell transplantation in autoimmune diseases in the peer-reviewed published medical literature. Such studies are necessary to determine whether stem cell transplantation improves clinical outcomes of autoimmune diseases. In addition, where stem cell transplantation is used with myeloablative chemotherapy of autoimmune diseases, the efficacy of stem cell transplantation in restoring hematopoiesis should be compared with administration of myelopoietic growth factors (granulocyte colony-stimulating facto [G-CSF] and granulocyte-macrophage colony-stimulating factor [GM-CSF]).

The position statement from a National Institute of Allergy and Infectious Diseases and National Cancer Institute-Sponsored International Workshop on "Feasibility of allogeneic hematopoietic stem cell transplantation for autoimmune disease" (Griffith et al, 2005) concluded that "[a] rationale clearly exists for exploring the therapeutic and curative potential of allogeneic HCT for severe autoimmune disease [systemic sclerosis, multiple sclerosis, systemic lupus erythematosus, rheumatoid arthritis, and autoimmune cytopenia]. Although safer allogeneic transplantation strategies have become available, experience is currently insufficient to allow reliable extrapolation of data on safety and risks from patients with malignancies to patients with autoimmune diseases. It is recommended that planning be initiated for clinical trials to generate safety and efficacy data for allogeneic HCT [hematopoietic stem cell transplant] in patients with severe autoimmune diseases".

Rice and colleagues (2013) stated that MS is incurable, but stem-cell therapy might offer valuable therapeutic potential. Efforts to develop stem-cell therapies for MS have been conventionally built on the principle of direct implantation of cells to replace oligodendrocytes, and therefore to regenerate myelin. Recent progress in understanding of disease processes in MS include observations that spontaneous myelin repair is far more widespread and successful than was previously believed, that loss of axons and neurons is more closely associated with progressive disability than is myelin loss, and that damage occurs diffusely throughout the central nervous system (CNS) in grey and white matter, not just in discrete, isolated patches or lesions. These findings had introduced new and serious challenges that stem-cell therapy needs to overcome; the practical challenges to achieve cell replacement alone are difficult enough, but, to be useful, cell therapy for MS must achieve substantially more than the replacement of lost oligodendrocytes. However, parallel advances in understanding of the reparative properties of stem cells -- including their distinct immunomodulatory and neuroprotective properties, interactions with resident or tissue-based stem cells, cell fusion, and neurotrophin elaboration -- offer renewed hope for development of cell-based therapies. Moreover, the authors stated that these advances suggested avenues for translation of this approach not only for MS, but also for other common neurological and neurodegenerative diseases.

Burt et al (2015) determined the association of non-myeloablative HSCT with neurological disability and other clinical outcomes in patients with MS. This study included case series of patients with relapsing-remitting MS (n = 123) or secondary-progressive MS (n = 28) (mean age of 36 years; range of 18 to 60 years; 85 women) treated at a single US institution between 2003 and 2014 and followed-up for 5 years. Final follow-up was completed in June 2014. Treatment with cyclophosphamide and alemtuzumab (22 patients) or

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

cyclophosphamide and thymoglobulin (129 patients) followed by infusion of un-manipulated peripheral blood stem cells. Primary end-point was reversal or progression of disability measured by change in the EDSS score of 1.0 or greater (score range of 0 to 10). Secondary outcomes included changes in the Neurologic Rating Scale (NRS) score of 10 or greater (score range of 0 to 100), Multiple Sclerosis Functional Composite (MSFC) score, quality-of-life Short Form 36 questionnaire scores, and T2 lesion volume on brain magnetic resonance imaging scan. Outcome analysis was available for 145 patients with a median follow-up of 2 years and a mean of 2.5 years. Scores from the EDSS improved significantly from a pre-transplant median of 4.0 to 3.0 (interquartile range [IQR]: 1.5 to 4.0; n = 82) at 2 years and to 2.5 (IQR, 1.9 to 4.5; n = 36) at 4 years ($p < 0.001$ at each assessment). There was significant improvement in disability (decrease in EDSS score of greater than or equal to 1.0) in 41 patients (50 %; 95 % CI: 39 % to 61 %) at 2 years and in 23 patients (64 %; 95 % CI: 46 % to 79 %) at 4 years. Four-year relapse-free survival was 80 % and progression-free survival (PFS) was 87 %. The NRS scores improved significantly from a pre-transplant median of 74 to 88.0 (IQR, 77.3 to 93.0; n = 78) at 2 years and to 87.5 (IQR, 75.0 to 93.8; n = 34) at 4 years ($p < 0.001$ at each assessment). The median MSFC scores were 0.38 (IQR, -0.01 to 0.64) at 2 years ($p < 0.001$) and 0.45 (0.04 to 0.60) at 4 years ($p = 0.02$). Total quality-of-life scores improved from a mean of 46 (95 % CI: 43 to 49) pre-transplant to 64 (95 % CI: 61 to 68) at a median follow-up of 2 years post-transplant (n = 132) ($p < 0.001$). There was a decrease in T2 lesion volume from a pre-transplant median of 8.57 cm3 (IQR, 2.78 to 22.08 cm3) to 5.74 cm3 (IQR, 1.88 to 14.45 cm3) ($p < 0.001$) at the last post-transplant assessment (mean follow-up of 27 months; n = 128). The authors concluded that among patients with relapsing-remitting MS, non-myeloablative HSCT was associated with improvement in neurological disability and other clinical outcomes. They stated that these preliminary findings from this uncontrolled study require confirmation in randomized trials.

The major drawbacks of this study were:

I. it was a single-center study, which may introduce the possibility of bias,
II. a number of patients were treated on a compassionate basis, rather than on the study protocol,
III. improvement in function in patients with MS may be in part a consequence of continuing recovery from earlier relapses,
IV. long-term follow-up (i.e., greater than or equal to 4 years) was not available for a substantial proportion of patients, and
V. this was an observational cohort study without a control group.

I. it was a single-center study, which may introduce the possibility of bias,
II. a number of patients were treated on a compassionate basis, rather than on the study protocol,
III. improvement in function in patients with MS may be in part a consequence of continuing recovery from earlier relapses,
IV. long-term follow-up (i.e., greater than or equal to 4 years) was not available for a substantial proportion of patients, and
V. this was an observational cohort study without a control group.

In an editorial that accompanied the afore-mentioned study, Hauser (2015) stated that

I. autologous HSCT does not appear to be effective against established progressive forms of MS and, absence of new data, additional trials of these protocols are probably not indicated for patients with progressive MS,
II. immunosuppressive regimens that include HSCT appear to be effective against relapsing-remitting form of MS , at least over several years of observation. However, it is by no means clear that the beneficial effects result from the infusion of stem cells rather than from the conditioning regimen. Studies that delineate the role of HSCT specifically in improving outcomes relative to conditioning regimens alone are needed before this therapy should be deployed outside of the clinical research setting,
III. the mechanism of action of autologous HSCT in MS needs to be clarified, and

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

IV. very long follow-up periods are needed to meaningfully assess the role of HSCT in the treatment of MS.

I. autologous HSCT does not appear to be effective against established progressive forms of MS and, absence of new data, additional trials of these protocols are probably not indicated for patients with progressive MS,
II. immunosuppressive regimens that include HSCT appear to be effective against relapsing-remitting form of MS , at least over several years of observation. However, it is by no means clear that the beneficial effects result from the infusion of stem cells rather than from the conditioning regimen. Studies that delineate the role of HSCT specifically in improving outcomes relative to conditioning regimens alone are needed before this therapy should be deployed outside of the clinical research setting,
III. the mechanism of action of autologous HSCT in MS needs to be clarified, and
IV. very long follow-up periods are needed to meaningfully assess the role of HSCT in the treatment of MS.

In a multi-center, phase II clinical trial, Mancardi et al (2015) examined the effect of intense immunosuppression followed by AHSCT versus mitoxantrone (MTX) on disease activity measured by MRI in patients with MS. Patients with secondary progressive or relapsing-remitting MS, with a documented increase in the last year on the EDSS, in spite of conventional therapy, and presence of 1 or more gadolinium-enhancing (Gd+) areas were included in this study. Patients were randomized to receive intense immunosuppression (mobilization with cyclophosphamide and filgrastim, conditioning with carmustine, cytosine-arabinoside, etoposide, melphalan, and anti-thymocyte globulin) followed by AHSCT or MTX 20 mg every month for 6 months. The primary end-point was the cumulative number of new T2 lesions in the 4 years following randomization. Secondary end-points were the cumulative number of Gd+ lesions, relapse rate, and disability progression. Safety and tolerability were also assessed; 21 patients were randomized and 17 had post-baseline evaluable MRI scans. Autologous HSCT reduced by 79 % the number of new T2 lesions as compared to MTX (rate ratio 0.21, $p = 0.00016$). It also reduced Gd+ lesions as well as the annualized relapse rate. No difference was found in the progression of disability. The authors concluded that intense immunosuppression followed by AHSCT is significantly superior to MTX in reducing MRI activity in severe cases of MS. They stated that these results strongly support further phase III studies with primary clinical end-points.

Stellmann et al (2015) noted that AHSCT is still not the standard treatment for highly inflammatory MS. Even though RCTs are lacking, predictors for treatment response have been established. Since 2007, 10 patients have received AHSCT in Hamburg. These researchers presented observational data from patients treated in Hamburg and a review of the literature. Descriptive statistics were used for evaluating the course of the EDSS as a measure for clinical outcome, MRI and neuropsychology; new gadolinium and T2-MRI uptake lesions per scan were compared. In addition, a systematic review of the currently available literature was performed. The Hamburg series can be divided in 2 groups -- one group including 4 patients with chronic progressive MS with low inflammatory activity (median EDSS = 6.25, 0.5 relapses per year, no gadolinium-enhancing lesions) and the other group including 6 patients with mild-to-moderate disability, relapses and inflammatory activity (median EDSS = 4.25, 1 relapse per year, 2 gadolinium-enhancing lesions). The median follow-up was 2.4 years. While the 1st group did not seem to benefit from AHSCT, an improvement in 5 out of 6 patients was observed in the 2nd group. New T2 lesions occurred within the first 6 months but gadolinium-enhancing lesions were not observed ($p < 0.05$). A systematic literature search identified a higher efficacy of AHSCT in younger, less disabled MS patients with inflammatory activity, similar to the findings from Hamburg. The authors concluded that cohort reports described AHSCT as a safe and efficient therapeutic option in highly inflammatory MS. The authors concluded that based on these data AHSCT appeared to be a reasonable option in selected patients with highly inflammatory MS; however a RCT is needed.

Li and colleagues (2016) noted that HCT has been reported for a long time and can be used for MS; however, available data consisted of small samples and gave confusing results. In a systematic review and meta-analysis, these investigators estimated the effects of HCT for adults with MS. They searched the database of CNKI, PubMed, Embase, WEB of SCIENCE and the Cochrane Center Register of Controlled Trials to find

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

initial studies and selected the appropriate researches included in the meta-analysis based on the inclusion and exclusion criteria; I2 was used to evaluate the heterogeneity and meta-regression was used for finding the source. Random effort model was performed to pool the data and funnel plot was drawn to determine publication bias; 8 single-arm clinical trials studies were included. The I2 value was 0.77 and 0.93, suggesting a heavy heterogeneity between studies. However, meta-regression analysis did not find the source of heterogeneity in which the publication country and follow up time were the influencing factors. Compared with baseline, the EDSS score of MS patients after HCT has a statistical decrease of 0.62 (95 % CI: 0.14 to 1.10) at the 12th month and 1.26 (95 % CI: 0.38 to 2.14) at the follow-up time ending point, respectively. The authors concluded that available evidence suggested that some clinical benefits of HCT when combined with immunotherapy on MS. However, as a consequence of wide CIs that were characteristics of small evidence bases, further investigations to provide enough baseline information according to the RCTs are needed for further analysis, such as subgroup analysis and meta-regression analysis.

Bakhuraysah and associates (2016) stated that HSCT is a treatment paradigm that has long been utilized for cancers of the blood and bone marrow, but has gained some traction as a treatment paradigm for MS. Success in the treatment of patients with this approach has been reported primarily when strict inclusion criteria were imposed that have eventuated a more precise understanding of MS pathophysiology, thereby governing trial design. Moreover, enhancing the yield and purity of hematopoietic stem cells during isolation along with the utility of appropriate conditioning agents has provided a clearer foundation for clinical translation studies. To support this approach, pre-clinical data derived from animal models of MS, experimental autoimmune encephalomyelitis, have provided clear identification of multipotent stem cells that can reconstitute the immune system to override the autoimmune attack of the CNS. These investigators discussed the rationale of HSCT to treat MS by providing the benefits and complications of the clinically relevant protocols, the varying graft types, and conditioning regimens. However, they emphasized that future trials based on HSCT should be focused on specific therapeutic strategies to target and limit ongoing neuro-degeneration and demyelination in progressive MS, in the hope that such treatment may serve a greater catchment of patient cohorts with potentially enhanced efficiency and lower toxicity. Despite these future ambitions, a proposed international multi-center, randomized clinical trial of HSCT should be governed by the best standard care of treatment, whereby MS patients are selected upon strict clinical course criteria and long-term follow-up studies of patients from international registries are imposed to advocate HSCT as a therapeutic option in the management of MS.

In a multi-center, single-group, phase II clinical trial, Atkins and co-workers (2016) examined if near-complete immuno-ablation followed by immune cell depleted AHSCT would result in long-term control of MS. These researchers enrolled patients with MS, aged 18 to 50 years with poor prognosis, ongoing disease activity, and an EDSS of 3.0 to 6.0. Autologous CD34 selected hemopoietic stem-cell grafts were collected after mobilization with cyclophosphamide and filgrastim. Immuno-ablation with busulfan, cyclophosphamide, and rabbit anti-thymocyte globulin was followed by AHSCT. The primary outcome was MS activity-free survival (events were clinical relapse, appearance of a new or Gd-enhancing lesion on MRI, and sustained progression of EDSS score). Between diagnosis and AHSCT, 24 patients had 167 clinical relapses over 140 patient-years with 188 Gd-enhancing lesions on 48 pre-AHSCT MRI scans. Median follow-up was 6.7 years (range of 3.9 to 12.7). The primary outcome, MS activity-free survival at 3 years after transplantation was 69.6 % (95 % CI: 46.6 to 84.2). With up to 13 years of follow-up after AHSCT, no relapses occurred and no Gd enhancing lesions or new T2 lesions were seen on 314 MRI sequential scans. The rate of brain atrophy decreased to that expected for healthy controls; 1 of 24 patients died of transplantation-related complications; 35 % of patients had a sustained improvement in their EDSS score. The authors described the first treatment to fully halt all detectable CNS inflammatory activity in patients with MS for a prolonged period in the absence of any ongoing disease-modifying drugs. Furthermore, many of the patients had substantial recovery of neurological function despite their disease's aggressive nature.

Nash and colleagues (2017) evaluated the safety, effectiveness, and durability of MS disease stabilization after high-dose immunosuppressive therapy (HDIT) and autologous HCT. High-Dose Immunosuppression and Autologous Transplantation for Multiple Sclerosis (HALT-MS) is a phase II clinical trial of HDIT/HCT for patients with relapsing-remitting MS (RRMS) who experienced relapses with disability progression (EDSS 3.0 to 5.5) while on MS disease-modifying therapy. The primary end-point was event-free survival

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

(EFS), defined as survival without death or disease activity from any one of the following: disability progression, relapse, or new lesions on MRI. Participants were evaluated through 5 years post-transplant. Toxicities were reported using the National Cancer Institute Common Terminology Criteria for adverse events (AEs). A total of 25 subjects were evaluated for transplant and 24 participants underwent HDIT/HCT. Median follow-up was 62 months (range of 12 to72); EFS was 69.2 % (90% CI: 50.2 to 82.1); PFS, clinical relapse-free survival, and MRI activity-free survival were 91.3 % (90 % CI: 74.7 % to 97.2 %), 86.9 % (90 % CI: 69.5 % to 94.7%), and 86.3 % (90 % CI: 68.1 % to 94.5 %), respectively; AE due to HDIT/HCT were consistent with expected toxicities and there were no significant late neurologic adverse effects noted. Improvements were noted in neurologic disability with a median change in EDSS of -0.5 (IQR -1.5 to 0.0; p = 0.001) among participants who survived and completed the study. The authors concluded that HDIT/HCT without maintenance therapy was effective for inducing long-term sustained remissions of active RRMS at 5 years. Moreover, they stated that prospective clinical trials comparing HDIT/HCT to other approaches are needed; HDIT/HCT may represent a potential therapeutic option for patients with RRMS who fail conventional immunotherapy. This study provided Class IV evidence that participants with RRMS experienced sustained remissions with toxicities as expected from HDIT/HCT.

Casanova and associates (2017) described the long-term results of myeloablative AHSCT in the treatment of patients with MS. Patients who failed to conventional therapies for MS underwent an approved protocol for AHSCT, which consisted of peripheral blood stem cell mobilization with cyclophosphamide and G-CSF, followed by a conditioning regimen of BCNU, etoposide, ara-C, melphalan IV, plus rabbit thymoglobulin. A total of 38 MS patients have been transplanted since 1999; 31 have been followed for more than 2 years (mean of 8.4 years). There were 22 RRMS patients and 9 SPMS patients; n death related to AHSCT. A total of 10 patients (32.3 %) had at least 1 relapse during post-AHSCT evolution, 6 patients in the RRMS group (27.2 %) and 4 in the SPMS group (44.4 %). After AHSCT, 7 patients (22.6 %) experienced progression of disability, all within SP form. By contrast, no patients with RRMS experienced worsening of disability after a median follow-up of 5.4 years, 60 % of them showed a sustained reduction in disability (SRD), defined as the improvement of 1.0 point in EDSS that sustained for 6 months (0.5 in cases of EDSS greater than or equal to 5.5). The only clinical variable that predicted a poor response to AHSCT was a high EDSS in the year before transplant. The authors concluded that AHSCT using the BEAM-ATG scheme is safe and effectives to control the aggressive forms of RRMS.

In a meta-analysis, Sormani and co-workers (2017) summarized the evidence on immuno-ablative therapy followed by AHSCT to manage severe and treatment-refractory MS. These investigators collected all the published studies of AHSCT in any form of MS from 1995 to 2016, carefully excluding reports that were updated in subsequent studies. End-points were transplant-related mortality (TRM), rate of disease progression, and no evidence of disease activity (NEDA) status. A weighted meta-regression based on a Poisson model was run, assessing whether there were study-specific characteristics with an effect on TRM and progression. A total of 15 studies including 764 transplanted patients were pooled in the meta-analysis. The pooled estimate of TRM was 2.1 % (95 % CI: 1.3 % to 3.4 %); TRM was higher in older studies (p = 0.014) and in studies with a lower proportion of patients with RRMS (p = 0.028). A higher baseline EDSS (p = 0.013) was also significantly associated with a higher TRM. Pooled rate of progression was 17.1 % at 2 years (95 % CI: 9.7 % to 24.5 %) and 23.3 % (95 % CI: 16.3 % to 31.8 %) at 5 years. Lower 2-year progression rate was significantly associated with higher proportions of patients with RRMS (p = 0.004). The pooled proportion of NEDA patients at 2 years was 83 % (range of 70 % to 92 %) and at 5 years was 67 % (range of 59 % to 70 %). The authors concluded that the emerging evidence on this therapeutic approach in MS indicated that the largest benefit/risk profile form this therapeutic approach can be obtained in patients with aggressive MS with a relapsing-remitting course and who have not yet accumulated a high level of disability.

In a large, multi-center, observational, retrospective cohort study, Murano and colleagues (2017) evaluated the long-term outcomes in patients who underwent autologous hematopoietic stem cell transplantation (AHSCT) for the treatment of MS. Eligibility criteria were receipt of AHSCT for the treatment of MS between January 1995 and December 2006 and the availability of a pre-specified minimum data set comprising the disease subtype at baseline; the EDSS score at baseline; information on the administered conditioning regimen and graft manipulation; and at least 1 follow-up visit or report after transplant. The last patient visit was on July 1, 2012. To avoid bias, all eligible patients were included in the analysis regardless

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

of their duration of follow-up. Data analysis was conducted from September 1, 2014 to April 27, 2015. Demographic, disease-related, and treatment-related exposures were considered variables of interest, including age, disease subtype, baseline EDSS score, number of previous disease-modifying treatments, and intensity of the conditioning regimen. The primary outcomes were MS PFS and OS. The probabilities of PFS and OS were calculated using Kaplan-Meier survival curves and multi-variable Cox proportional hazards regression analysis models. Valid data were obtained from 25 centers in 13 countries for 281 evaluable patients, with median follow-up of 6.6 years (range of 0.2 to 16); 78 % (218 of 281) of patients had progressive forms of MS. The median EDSS score before mobilization of peripheral blood stem cells was 6.5 (range of 1.5 to 9); 8 deaths (2.8 %; 95 % CI: 1.0 % to 4.9 %) were reported within 100 days of transplant and were considered transplant-related mortality. The 5-year probability of PFS as assessed by the EDSS score was 46 % (95 % CI: 42 % to 54 %), and OS was 93 % (95 % CI: 89 % to 96 %) at 5 years. Factors associated with neurological progression after transplant were older age (hazard ratio [HR], 1.03; 95 % CI: 1.00 to 1.05), progressive versus relapsing form of MS (HR, 2.33; 95 % CI: 1.27 to 4.28), and more than 2 previous disease-modifying therapies (HR, 1.65; 95 % CI: 1.10 to 2.47). Higher baseline EDSS score was associated with worse OS (HR, 2.03; 95 % CI: 1.40 to 2.95). The authors concluded that in this observational study of patients with MS treated with AHSCT, almost 50 % of them remained free from neurological progression for 5 years after transplant. Younger age, relapsing form of MS, fewer prior immunotherapies, and lower baseline EDSS score were factors associated with better outcomes. They stated that the findings of this study support the rationale for further randomized clinical trials of AHSCT for the treatment of MS.

The authors stated that the main drawback of this study was its partially retrospective nature. Although some of the data were obtained retrospectively from clinical records, these investigators took many steps to optimize the analysis. As with most database studies, the reported outcomes mirrored the practice for MS treatment in many countries. The raters of EDSS assessments were not masked to clinical information, and assessments were not systematically performed for the duration of follow-up in every patient. Therefore, these researchers limited the analysis of PFS to the large 85.1 % (239 of 281) subset of patients who had yearly EDSS assessments after transplant. The number of patients with enough data points for the different analyses was variable and sometimes low, which reduced statistical power. In a retrospective study, incomplete reporting and loss to follow-up may result in under-estimating the frequency of late adverse events (AEs). Another limitation was that, although the analysis included 57.0 % (281 of 493) of the transplants registered with the CIBMTR and the EBMT during the study period, more than 1/3 of the activity was not captured by this study. However, the reason for the 78.3 % (166 of 212) of unavailable cases was that the centers where the patients were treated declined to participate in the study; 74 % (43 of 58) of centers that did not join the study had performed fewer than 3 transplants, and lack of incentive for the clinicians to contribute few cases to a large study was stated in many centers' responses. Based on this information, the authors do not expect that the unavailability of those cases could represent a significant source of bias.

Burman and colleagues (2017) noted that AHSCT is a promising therapy for MS, which has mainly been used in adults. These researchers examined the effectiveness and AEs of AHSCT in the treatment of children with MS using data from the European Society for Blood and Marrow Transplantation registry. A total of 21 patients with a median follow-up time of 2.8 years could be identified; PFS at 3 years was 100 %, 16 patients improved in EDSS score and only 2 patients experienced a clinical relapse. The procedure was generally well-tolerated and only 2instances of severe transplant-related toxicity were recorded. There was no treatment-related mortality, although 1 patient needed intensive care. The authors concluded that AHSCT may be a therapeutic option for children with disease that does not respond to standard care.

In a randomized clinical trial, Burt and colleagues (2019) compared the effect of non-myeloablative HSCT versus disease-modifying therapy (DMT) on disease progression in patients with RRMS. Between September 20, 2005, and July 7, 2016, a total of 110 patients with RRMS, at least 2 relapses while receiving DMT in the prior year, and an EDSS (score range of 0 to 10 [10 = worst neurologic disability]) score of 2.0 to 6.0 were randomized at 4 US, European, and South American centers. Final follow-up occurred in January 2018 and database lock in February 2018. Patients were randomized to receive HSCT along with cyclophosphamide (200 mg/kg) and anti-thymocyte globulin (6 mg/kg) (n = 55) or DMT of higher efficacy or a different class than DMT taken during the previous year (n = 55). The primary end-point was disease progression, defined as an EDSS score increase after at least 1 year of 1.0 point or more (minimal clinically

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

important difference, 0.5) on 2 evaluations 6 months apart, with differences in time to progression estimated as HRs. Among 110 randomized patients (73 [66 %] women; mean age of 36 [SD, 8.6] years), 103 remained in the trial, with 98 evaluated at 1 year and 23 evaluated yearly for 5 years (median follow-up of 2 years; mean of 2.8 years). Disease progression occurred in 3 patients in the HSCT group and 34 patients in the DMT group. Median time-to-progression could not be calculated in the HSCT group because of too few events; it was 24 months (IQR, 18 to 48 months) in the DMT group (HR, 0.07; 95 % CI: 0.02 to 0.24; $p < 0.001$). During the 1st year, mean EDSS scores decreased (improved) from 3.38 to 2.36 in the HSCT group and increased (worsened) from 3.31 to 3.98 in the DMT group (between-group mean difference, -1.7; 95 % CI: -2.03 to -1.29; $p < 0.001$). There were no deaths and no patients who received HSCT developed non-hematopoietic grade 4 toxicities (such as myocardial infarction, sepsis, or other disabling or potential life-threatening events). The authors concluded that in this preliminary study of patients with RRMS, non-myeloablative HSCT, compared with DMT, resulted in prolonged time to disease progression. Moreover, these researchers stated that further research is needed to replicate these findings and to assess long-term outcomes and safety.

The authors stated that this study had several drawbacks. First, a relatively small number of patients were treated compared with pharmaceutical sponsored trials, and the relatively small sample size resulted in small numbers of patients available to assess longer-term outcomes. The number of patients needed in randomized trials with active comparators depends on the known or expected treatment effect of the 2 treatments. Previous DMT trials for RRMS needed a fairly large number of patients to show the superiority of a new DMT compared with interferons or glatiramer acetate. Because HSCT efficacy was assumed to be superior to DMT, the MIST trial was designed with a smaller number of patients. Second, the study design allowed patients in the DMT group in whom that treatment failed to cross-over to receive HSCT, which also limited the ability to collect follow-up data for patients receiving DMT and to assess longer-term secondary outcomes. Because of ethical concerns of treatment equipoise between the 2 groups, the cross-over option was included for patients whose EDSS worsened with continued DMT treatment. The cross-over prevented comparison of the HSCT and DMT groups after 1 year; but did not affect the primary end-point of time to progression or the end-points of time to first relapse or no evidence of disease activity, and allowed completion of the study. Even with the provision for cross-over, 4 patients in the DMT group left the study to receive HSCT at other sites. Cross-over did allow patients in whom DMT continued to fail to be their own control, demonstrating, despite continued failure of DMT, marked improvement after HSCT. Third, alemtuzumab and ocrelizumab were excluded from use in the DMT group. Ocrelizumab was not included as DMT because the study completed enrollment in 2016 and ocrelizumab was not FDA licensed until 2017. Alemtuzumab was FDA licensed in 2014 but was excluded because prolonged alemtuzumab-induced lymphopenias and secondary autoimmune disorders could contribute to or cause post-HSCT infections or autoimmune diseases in the cross-over group. In comparison with this study, which allowed multiple DMTs in the control group, controls for pharmaceutical DMT trials have been limited to placebo, interferon, and glatiramer acetate. Fourth, although the evaluating neurologist for EDSS and NRS was masked to treatment assignment, the physician who recorded relapses was not masked. Fifth, although patients at all sites had significant improvement in EDSS after HSCT and the study was numerically weighted toward the US site, 1 site enrolled patients with lower disability scores, and at that site the primary outcome of progression between HSCT and DMT groups was less pronounced. However, it would be anticipated that the rate of progression would be slower in less disabled patients.

In an editorial that accompanied the afore-mentioned study, Atkins (2019) stated that several questions remain to be addressed: When in the course of MS is HSCT most optimally used? How many disease-modifying therapy (DMT) should fail in a patient before considering HSCT? What is the optimal transplant conditioning regimen that balances toxicity and efficacy in controlling MS? And should HSCT be more liberally applied, for instance, for patients with less frequent relapses? The editorialist stated that careful ongoing study, through clinical trials and registry-based databases, are needed to optimize the timing and sequence of DMT and HSCT, as the use of HSCT moves into the realm of potential treatment options for patients with MS.

Cohen et al (2019) noted that MS is a chronic, disabling, immune-mediated, demyelinating and degenerative disease of the CNS. Approved disease-modifying therapies may be incompletely effective in some patients with highly active relapsing disease and high risk of disability. The use of immune-ablative or myeloablative

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

therapy followed by autologous hematopoietic cell transplantation (AHCT) has been examined in retrospective studies, clinical trials, and meta-analyses/systematic reviews as an approach to address this unmet clinical need. On behalf of the American Society for Blood and Bone Marrow Transplantation (ASBMT), a panel of experts in AHCT and MS convened to review available evidence and make recommendations on MS as an indication for AHCT. A review of recent literature identified 8 retrospective studies, 8 clinical trials, and 3 meta-analyses/systematic reviews. In aggregate, these studies showed that AHCT was a safe and effective treatment for active relapsing forms of MS to prevent clinical relapse, MRI-detectable lesion activity, and worsening disability and to reverse disability without unexpected AEs. Based on the available evidence, the ASBMT recommended that treatment-refractory relapsing MS with high risk of future disability be considered a "standard of care, clinical evidence available" indication for AHCT. Collaboration of neurologists with expertise in treating MS and transplantation physicians with experience performing AHCT for autoimmune disease is crucial for ensuring appropriate patient selection and optimizing transplantation procedures to improve patient outcomes. Transplantation centers in Canada and the U.S. were strongly encouraged to report baseline and outcomes data on patients receiving AHCT for MS to the Center for International Blood and Marrow Transplant Research. The authors concluded that the retrospective studies and clinical trials of AHCT discussed in this document differed in design, study population, and transplantation protocol, and only 2 studies included a control group. Typically, the AHCT studies enrolled patients with more active or severe disease compared with the studies of conventional DMTs. Despite these limitations, it is apparent that after AHCT, many patients with MS experienced rapid, complete, and durable control of inflammatory disease activity, with resultant improvement in long-term clinical outcomes. Where data for similar MS populations are available, outcomes with AHCT appeared to be superior compared with reports for conventional disease-modifying therapies. Moreover, these investigators stated that this document is not a treatment guideline, but is intended to provide guidance to physicians, patients, payers, policy makers, and other stakeholders on coverage decisions and the appropriate use of this procedure for MS.

In a retrospective, cohort, open-label study, Nicholas et al (2021) examined outcomes in patients with MS (PwMS) treated with AHSCT in a real-world setting. This trial included PwMS treated with AHSCT at 2 centers in London, consecutively between 2012 and 2019 who had greater than or equal to 6 months of follow-up or died at any time. Primary outcomes were survival free of MS relapses, MRI new lesions, and worsening of EDSS score; AEs rates were also examined. The cohort includes 120 PwMS; 52 % had progressive MS (primary or secondary) and 48 % had relapsing-remitting MS. At baseline, the median EDSS score was 6.0; 90 % of the evaluable cases showed MRI activity in the 12 months preceding AHSCT. Median follow-up after AHSCT was 21 months (range of 6 to 85 months). MS relapse-free survival was 93 % at 2 years and 87 % at 4 years following AHSCT. No new MRI lesions were detected in 90 % of subjects at 2 years and in 85 % at 4 years. EDSS score PFS was 75 % at 2 years and 65 % at 4 years. Epstein-Barr virus (EBV) re-activation and monoclonal para-proteinemia were associated with worse PFS. There were 3 transplantation-related deaths within 100 days (2.5 %), all after fluid over-load and cardiac or respiratory failure. The authors concluded that effectiveness outcomes of AHSCT in this real-world cohort were similar to those reported in more stringently selected clinical trial populations, although the risks may be higher. This study was rated Class IV because of the uncontrolled, open-label design.

Patti et al (2022) stated that MS is a chronic, inflammatory and immune-mediated disease of the CNS, commonly affecting young adults and potentially associated with life-long disability. About 14 disease-modifying treatments (DMTs) are currently approved for the treatment of MS. However, despite the use of highly effective therapies, some patients exhibit a highly active disease with an aggressive course from onset and a higher risk of long-term disability accrual. In the past few years, several retrospective studies, clinical trials, meta-analyses and systematic reviews have examined the use of AHSCT as a possible therapeutic option to address this unmet clinical need. These studies demonstrated that AHSCT is a highly effective and relatively safe therapeutic option for the treatment of highly active MS. In particular, I the past several years, the amount of evidence has grown, with significant improvements in the development of patient selection criteria, choice of the most suitable transplant technique and clinical experience. The authors presented the findings of 6 patients who received AHSCT in their MS center and these investigators systematically reviewed recent evidence regarding the long-term safety and effectiveness of AHSCT and the placement of AHSCT in the rapidly evolving therapeutic armamentarium for MS. Moreover, these researchers stated that definitive evidence from new and ongoing clinical trials is awaited from ongoing RCTs in order to endorse

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

AHSCT as “standard of care, clinical evidence available” for the treatment of aggressive and refractory relapsing MS patients.

Wilson et al (2022) noted that AHSCT is gaining traction as a valuable therapeutic option for patients affected by severe MS, especially the relapsing-remitting form. These researchers described the available evidence in terms of clinical trials, observational and retrospective studies, as well as immune reconstitution following transplantation, with a focus on the conditioning regimens used for transplantation. The evidence base predominantly consists of non-randomized, uncontrolled clinical trials or data from retrospective or observational cohorts (i.e., very few randomized or controlled trials). Most often, intermediate-intensity conditioning regimens were used, with promising results from both myeloablative and lympho-ablative strategies, as well as from regimens that were of low- and high-intensity. Effectiveness of transplantation, which is likely secondary to immune reconstitution and restored immune tolerance, is, thus, not clearly dependent on the intensity of the conditioning regimen. However, the conditioning regimen may well influence the immune response to transplantation. Heterogeneity of conditioning regimens among studies hinders synthesis of the articles examining post-AHSCT immune system changes. Factors associated with better outcomes were lower Kurtzke EDSS, RRMS, younger age, and shorter disease duration at baseline, which supports the guidance for patient selection proposed by the EBMT. Interestingly, promising outcomes were described for patients with SPMS by some studies, which may be worth taking into account when considering therapeutic options for patients with active, progressive disease. These researchers stated that RCTs are urgently needed to generate data that clearly indicates AHSCT is superior to other, less aggressive, therapeutic options available, with 3 clinical trials comparing AHSCT to either alemtuzumab or the best-available therapeutic option currently in progress. It will be of interest to note if these trials also observe an increased prevalence of autoimmune disease in their alemtuzumab populations and indeed in the cohort of patients receiving AHSCT, as this appeared to be an important outcome parameter that affects a significant proportion of patients in the long-term. Why this occurs is certainly a research area of interest, as is exactly how immune reconstitution would lead to restored immune tolerance in the patients who benefit from transplantation.

Hendrawan et al (2022) stated that AHSCT is a therapeutic option for hematological malignancies, such as non-Hodgkin's lymphoma (NHL), and more recently, for autoimmune diseases, such as treatment-refractory MS. The immunological mechanisms underlying remission in MS patients following AHSCT likely involve an anti-inflammatory shift in the milieu of circulating cytokines. These researchers hypothesized that immunological tolerance in MS patients post-AHSCT is reflected by an increase in anti-inflammatory cytokines and a suppression of pro-inflammatory cytokines in the patient blood. They examined this hypothesis using a multiplex-ELISA assay to compare the concentrations of secreted cytokine in the peripheral blood of MS patients and NHL patients undergoing AHSCT. In MS patients, these investigators detected significant reductions in pro-inflammatory T helper (Th)17 cytokines interleukin (IL)-17, IL-23, IL-1β, and IL-21, and Th1 cytokines interferon (IFN)γ and IL-12p70 in MS patients from day 8 to 24 months post-AHSCT. These changes were not observed in the NHL patients despite similar pre-conditioning treatment for AHSCT. Some pro-inflammatory cytokines demonstrated similar trends in both cohorts, such as IL-8 and tumor necrosis factor (TNF)-α, indicating a probable treatment-related AHSCT response. Anti-inflammatory cytokines (IL-10, IL-4, and IL-2) were only transiently reduced post-AHSCT, with only IL-10 exhibiting a significant surge at day 14 post-AHSCT. MS patients who relapsed post-AHSCT exhibited significantly elevated levels of IL-17 at 12 months post-AHSCT, unlike non-relapse patients who displayed sustained suppression of Th17 cytokines at all post-AHSCT time-points up to 24 months. The authors concluded that these findings suggested that suppression of Th17 cytokines was essential for the induction of long-term remission in MS patients following AHSCT. Moreover, these researchers stated that future studies are needed to examine the relationship between these post-AHSCT cytokine profiles to the reconstitution of immune cell populations to further elucidate the mechanisms of immunological tolerance regeneration in MS patients following transplantation.

The authors stated that the drawbacks of this trial were the relatively small numbers of patient cohorts and controls, and the lack of anti-thymocyte globulin (ATG) and use of more intensive immune chemotherapy in the control group. Furthermore, this study did not include data for B/T cell immune reconstitution following AHSCT. Nevertheless, this study represented the 1st significant analysis of cytokine expression in MS patients undergoing AHSCT using BEAM (carmustine, etoposide, cytarabine, and melphalan)/ATG.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Boffa et al (2023) stated that uncontrolled evidence suggested that AHSCT can be effective in individuals with active SPMS. These investigators compared the effect of AHSCT with that of other anti-inflammatory DMTs on long-term disability worsening in active SPMS. They collected data from the Italian Bone Marrow Transplantation Study Group and the Italian Multiple Sclerosis Register. Patients were considered eligible if treatment had been started after the diagnosis of SPMS. Disability worsening was assessed by the cumulative proportion of patients with a 6-month confirmed disability progression (CDP) according to the EDSS score. Key secondary endpoints were the EDSS time trend after treatment start and the prevalence of disability improvement over time. Time to first CDP was assessed by means of proportional hazard Cox regression models. A linear mixed model with a time × treatment group interaction was used to assess the longitudinal EDSS time trends. Prevalence of improvement was estimated using a modified Kaplan-Meier estimator and compared between groups by boot-strapping the area under the curve (AUC). A total of 79 AHSCT-treated patients and 1,975 patients treated with other DMTs (beta interferons, azathioprine, glatiramer-acetate, mitoxantrone, fingolimod, natalizumab, methotrexate, teriflunomide, cyclophosphamide, dimethyl fumarate, and alemtuzumab) were matched to reduce treatment selection bias using propensity score and overlap weighting approaches. Time to first CDP was significantly longer in transplanted patients (HR = 0.50; 95 % CI: 0.31 to 0.81; $p = 0.005$), with 61.7 % of transplanted patients free from CPD at 5 years. Accordingly, EDSS time trend over 10 years was higher in patients treated with other DMTs than in AHSCT-treated patients (+0.157 EDSS points per year compared with -0.013 EDSS points per year; interaction $p < 0.001$). Patients who underwent AHSCT were more likely to experience a sustained disability improvement: 34.7 % of patients maintained an improvement (a lower EDSS than baseline) 3 years after transplant versus 4.6 % of patients treated by other DMTs ($p < 0.001$). The authors concluded that the use of AHSCT in individuals with active SPMS was associated with a slowing of disability progression and a higher likelihood of disability improvement compared with standard immunotherapy. This study provided Class III evidence that AHSCT prolonged the time to CDP compared with other DMTs. Moreover, these researchers stated that it is important to note that because this cohort was composed of relatively young patients with clinical activity during SPMS, the findings of this study could not be applicable to patients with SPMS without signs of inflammatory disease activity.

Medina et al (2024) provided an overview of autoimmune disorders of the CNS, specifically those caused by demyelination. These investigators examined new research regarding potential therapeutic interventions, especially those aimed at inducing re-myelination, which is a detailed process, involving many cell types -- oligodendrocyte precursor cells (OPCs), astrocytes, and microglia-and both the innate and adaptive immune systems. The discussion of this process included the differentiation potential of neural stem cells, the function of adult OPCs, and the impact of molecular mediators on myelin repair. Emerging therapies were also explored, with mechanisms of action including the induction of OPC differentiation, the transplantation of mesenchymal stem cells, and the use of molecular mediators. Furthermore, these researchers discussed current medical advancements in relation to many myelin-related disorders, including MS, optic neuritis, neuromyelitis optica spectrum disorder, myelin oligodendrocyte glycoprotein antibody-associated disease, transverse myelitis, and acute disseminated encephalomyelitis. Beyond these emerging systemic therapies, the authors also introduced the dimethyl fumarate/silk fibroin nerve conduit and its potential role in the treatment of peripheral nerve injuries. Despite these afore-mentioned scientific advancements, these researchers maintained the need for ongoing research to deepen the understanding of demyelinating diseases and advance therapeutic strategies that enhance affected patients' QOL. These investigators stated that there have been experimental approaches to cell-based therapies, including mesenchymal stem cells and OPCs derived from bone marrow and other tissues. There remain many unresolved barriers to implementing the transplant-based approach to MS, such as the mode of delivery, tumor-facing potential, and requirements for immune suppression. Ultimately, there is a need for more advanced clinical trials. The role of immune-ablation and autologous hematopoietic stem cell transplantation is being investigated.

Lira-Lara et al (2025) noted that AHSCT is a viable therapeutic approach in patients with autoimmune diseases. Since June 2015, these researchers have autografted on an outpatient basis 1,700 AHSCT patients. They examined the salient features of early post-AHSCT complications when carried out in the outpatient setting. The primary end-points were early morbimortality-free survival (MFS) and OS, whereas secondary end-points entailed hospital admissions, neutropenic fever, MS flare-up, pneumothorax, hyponatremia as well as myocarditis. A total of 1,700 consecutive AHSCT recipients were analyzed: 1,667 with MS, 29 with CIDP, and 4 with other autoimmune diseases. A total of 1,643 (96.6 %) grafts were fully completed in the

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

outpatient setting. The 30-day MFS and 30-day OS were 87.7 % and 99.8 %, respectively. The 30-day MFS has increased from 94.9 % in the first 5 years to 98.2 % in the last 5 years (p = 0.0002). The 28-day mortality was 0.17 %, whereas the 28-day morbidity was 3.3 %. The rate of early complications decreased over time, most likely reflecting a learning curve effect. The authors concluded that these findings supported that employing this method was safe in the short-term. However, as this has been carried out in a "trial" setting, further investigations are needed.

Furthermore, an UpToDate review on "Overview of disease-modifying therapies for multiple sclerosis" (Olek and Mowry, 2025) states that "Relapsing forms of MS … These and other reports illustrate the potential benefits and perils of HSCT. More long-term data from randomized controlled trials are needed to assess the efficacy and safety of this intervention for the treatment of highly active RRMS …. Hematopoietic and mesenchymal stem cell therapy are promising investigational therapies for MS, but long-term data from randomized controlled trials are needed to assess the efficacy and safety of these therapies".

Polycythemia Vera

Polycythemia vera (PV) in children and adolescents is very rare. Data on clinical and laboratory evaluations as well as on treatment modalities are sparse. Cario and associates (2009) reported the long-term clinical course of a PV patient first diagnosed more than 40 years ago at age 12. In addition, after a systematic review of the scientific medical literature, clinical and hematological data of 36 patients (17 males and 19 females) from 25 previous reports were summarized. Three patients developed PV following antecedent hematological malignancies; Budd-Chiari syndrome was diagnosed in 7 patients indicating a particular risk of young patients of developing this disorder. One patient presented with ischemic stroke, 1 patient with gangrene, and 3 patients with severe hemorrhage, 3 patients died from disease-related complications. Hematocrit levels and platelet counts were not correlated with disease severity. Leukocytosis greater than 15 x 10(9)/L was present in 9/35 patients and associated with a thromboembolic or hemorrhagic complication in 7 patients. The few available data on molecular genetics and endogenous erythroid colony growth indicate changes comparable to those detectable in adult patients. Treatment varied enormously. It included aspirin, phlebotomy, hydroxycarbamide, busulfan, melphalan, pyrimethamine, and interferon-alpha. Two patients successfully underwent stem cell transplantation. The authors stated that it is currently impossible to treat an individual pediatric PV patient with an evidence-based regimen.

Tefferi and Vainchenker (2011) updated oncologists on pathogenesis, contemporary diagnosis, risk stratification, and treatment strategies in BCR-ABL1-negative myeloproliferative neoplasms, including PV, essential thrombocythemia (ET), and primary myelofibrosis (PMF). Recent literature was reviewed and interpreted in the context of the authors' own experience and expertise. Pathogenetic mechanisms in PV, ET, and PMF include stem cell-derived clonal myeloproliferation and secondary stromal changes in the bone marrow and spleen. Most patients carry an activating JAK2 or MPL mutation and a smaller subset also harbors LNK, CBL, TET2, ASXL1, IDH, IKZF1, or EZH2 mutations; the precise pathogenetic contribution of these mutations is under investigation. JAK2 mutation analysis is now a formal component of diagnostic criteria for PV, ET, and PMF, but its prognostic utility is limited. Life expectancy in the majority of patients with PV or ET is near-normal and disease complications are effectively (and safely) managed by treatment with low-dose aspirin, phlebotomy, or hydroxyurea. In PMF, survival and quality of life are significantly worse and current therapy is inadequate. In ET and PV, controlled studies are needed to show added value and justify the risk of unknown long-term health effects associated with non-conventional therapeutic approaches (e.g., interferon-alfa). The authors stated that the unmet need for treatment in PMF dictates a different approach for assessing the therapeutic value of new drugs (e.g., JAK inhibitors, pomalidomide) or allogeneic stem-cell transplantation.

Furthermore, Cancer Care Ontario's guidelines on the management of malignant thrombocytosis in Philadelphia chromosome-negative myeloproliferative disease (specifically ET or PV) (Matthews et al, 2008) as well as the National Health, Lung and Blood Institute's Diseases and Conditions Index on polycythemia vera (2011) do not list stem cell transplant as a therapeutic option.

Recessive Dystrophic Epidermolysis Bullosa

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Recessive dystrophic epidermolysis bullosa is an incurable, often fatal mucocutaneous blistering disease caused by mutations in COL7A1, the gene encoding type VII collagen (C7). On the basis of pre-clinical data showing biochemical correction and prolonged survival in col7 −/− mice, Wagner et al (2010) hypothesized that allogeneic marrow contains stem cells capable of ameliorating the manifestations of recessive dystrophic epidermolysis bullosa in humans. Between October 2007 and August 2009, these researchers treated 7 children who had recessive dystrophic epidermolysis bullosa with immuno-myeloablative chemotherapy and allogeneic stem-cell transplantation. They assessed C7 expression by means of immunofluorescence staining and used transmission electron microscopy to visualize anchoring fibrils. These investigators measured chimerism by means of competitive polymerase-chain-reaction assay, and documented blister formation and wound healing with the use of digital photography. One patient died of cardiomyopathy before transplantation. Of the remaining 6 patients, 1 had severe regimen-related cutaneous toxicity, with all having improved wound healing and a reduction in blister formation between 30 and 130 days after transplantation. These researchers observed increased C7 deposition at the dermal-epidermal junction in 5 of the 6 recipients, albeit without normalization of anchoring fibrils. Five recipients were alive 130 to 799 days after transplantation; 1 died at 183 days as a consequence of graft rejection and infection. The 6 recipients had substantial proportions of donor cells in the skin, and none had detectable anti-C7 antibodies. The authors concluded that increased C7 deposition and a sustained presence of donor cells were found in the skin of children with recessive dystrophic epidermolysis bullosa after allogeneic bone marrow transplantation. They stated that further studies (with more objective methods to evaluate the frequency of blistering, larger patient cohorts, and longer follow-up periods) are needed to assess the long-term risks and benefits of such therapy in patients with this disorder.

Retinitis Pigmentosa

Li et al (2012) stated that the U.S. Food and Drug Administration recently approved phase I/II clinical trials for embryonic stem (ES) cell-based retinal pigmented epithelium (RPE) transplantation, but this allograft transplantation requires life-long immunosuppressive therapy. Autografts from patient-specific induced pluripotent stem (iPS) cells offer an alternative solution to this problem. However, more data are required to establish the safety and effectiveness of iPS transplantation in animal models before moving iPS therapy into clinical trials. This study examined the effectiveness of iPS transplantation in restoring functional vision in Rpe65(rd12)/Rpe65(rd12) mice, a clinically relevant model of RP. Human iPS cells were differentiated into morphologically and functionally RPE-like tissue. Quantitative real-time polymerase chain reaction (RT-PCR) and immunoblots confirmed RPE fate. The iPS-derived RPE cells were injected into the sub-retinal space of Rpe65(rd12)/Rpe65(rd12) mice at 2 days postnatally. After transplantation, the long-term surviving iPS-derived RPE graft co-localized with the host native RPE cells and assimilated into the host retina without disruption. None of the mice receiving transplants developed tumors over their life-times. Furthermore, electroretinography demonstrated improved visual function in recipients over the life-time of this RP mouse model. The authors concluded that this study provided the first direct evidence of functional recovery in a clinically relevant model of retinal degeneration using iPS transplantation and supports the feasibility of autologous iPS cell transplantation for retinal and macular degenerations featuring significant RPE loss.

Rheumatoid Diseases

Liu et al (2014) noted that the clinical management of autoimmune rheumatic diseases (ARD) has undergone significant changes in the last few decades, leading to remarkable improvements in clinical outcomes of many patients with mild to moderate ARD. On the other hand, severe refractory ARD patients often have high morbidity and mortality. Extensive basic research and clinical evidence has opened the door to new encouraging perspectives, such as the establishment of a role of HSCT in the strategic management of ARD. Given the great heterogeneity of ARD, it is difficult to assign an optimal HSCT regimen to all ARD patients. The authors concluded that HSCT remains a challenging mode of therapy in ARD patients from the standpoints of both safety and effectiveness. As the clinical data of HSCT in ARD increases and as the understanding of stem cell biology and the downstream effects on the immune system increases, the future is promising for the development of optimal personalized HSCT regimens in ARD.

Systemic Lupus Sclerosus

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Loh et al (2007) stated that patients with cardiac dysfunction may be at increased risk of cardiac toxicity when undergoing HCT, which may preclude them from receiving this therapy. Moreover, cardiac dysfunction is common in SLE patients. While autologous HCT (auto-HCT) has been performed increasingly for SLE, its impact on cardiac function has not previously been evaluated. Thus, these investigators performed a retrospective analysis of SLE patients who had undergone auto-HCT to determine the prevalence of significant cardiac involvement, and the impact of transplantation on this. The records of 55 patients were reviewed, of which 13 were found to have abnormal cardiac findings on pre-transplant 2-dimensional echocardiography or multi-gated acquisition scan: impaired left ventricular ejection fraction (LVEF) (n = 6), pulmonary hypertension (n = 5), mitral valve dysfunction (n = 3) and large pericardial effusion (n = 1). At a median follow-up of 24 months (8 to 105 months), there were no transplant-related or cardiac deaths. With transplant-induced disease remission, all patients with impaired LVEF remained stable or improved; while 3 with symptomatic mitral valve disease similarly improved. Elevated pulmonary pressures paralleled activity of underlying lupus. These data suggest that auto-HCT is feasible in selected patients with lupus-related cardiac dysfunction, and with control of disease activity, may improve. The authors stated that the findings of this study warrant a prospective study with a larger cohort of SLE patients. It is hoped that with further investigation, the selection of which patients with cardiac abnormalities would benefit from auto-HCT and justify the potential risk will be made clearer.

Song and colleagues (2011) examined the effectiveness and toxicity of autologous stem cell transplantation (auto-SCT) in patients with SLE (n = 17). Peripheral blood stem cells were mobilized with cyclophosphamide (Cy) and granulocyte colony-stimulating factor. After a conditioning regimen of Cy and anti-thymocyte globulin, stem cells was re-infused. The probabilities of overall survival (OS) and progression-free survival (PFS) were used to assess the efficacy and adverse experiences, to detect the toxicities of the treatment. The median follow-up time was 89 months (range of 33 to 110). Probabilities of 7-year OS and PFS were 82.4 % +/- 9.2 % and 64.7 % +/- 11.6 %, respectively. The principal adverse events included allergy, infection, elevation of liver enzymes, bone pain, and heart failure. Two patients died due to severe pneumonia and heart failure at 33 and 64 months after transplantation, respectively. The authors concluded that their 7-year follow-up results suggested that auto-SCT seemed beneficial for SLE patients. Moreover, they stated that recruitment of more patients into multi-center, randomized, comparative studies versus conventional treatment of SLE is needed to evaluate the safety and effectiveness of auto-SCT.

Thrombotic Thrombocytopenic Purpura

UpToDate reviews on “Treatment and prognosis of thrombotic thrombocytopenic purpura-hemolytic uremic syndromes in adults” (Kaplan and George, 2013) does not mention the use of transplantation as a therapeutic option.

Ulcerative Colitis

UpToDate reviews on “Management of severe ulcerative colitis” (Peppercorn and Farrell, 2013) and “Treatment of ulcerative colitis in children and adolescents” (Bousvaros et al, 2013) do not mention the use of transplantation as a therapeutic option.

Mesenchymal Stem Cells

Wu and colleagues (2013) stated that mesenchymal stem cells (MSCs) have been shown to be effective in the management of graft-versus-host disease (GVHD) due to their immunomodulatory effects. In addition to prevention and treatment of GVHD, many studies have demonstrated that MSCs can promote hematopoietic engraftment, accelerate lymphocyte recovery, reduce the risk of graft failure, and repair tissue damage in patients receiving HSCT. Bone marrow (BM) has been considered as the traditional source of MSCs, and most of the knowledge concerning MSCs comes from BM studies. However, BM-derived MSCs have several limitations for their clinical application. Fetal-type MSCs can be isolated easier and proliferate faster in-vitro as well as possessing a lower immunogenicity. Therefore, fetal-type MSCs, such as umbilical cord-derived MSCs, represent an excellent alternative source of MSCs. Mesenchymal stem cells play multiple important roles in HSCT. Nevertheless, several issues regarding their clinical application remain to be

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

discussed, including the safety of use in humans, the available sources and the convenience of obtaining MSCs, the quality control of in vitro-cultured MSCs and the appropriate cell passages, the optimum cell dose, and the optimum number of infusions. Furthermore, it is important to evaluate whether the rates of cancer relapse and infections increase when using MSCs for GVHD. The authors concluded that there are still many questions regarding the clinical application of MSCs to HSCT that need to be answered, and further studies are needed.

Batsali et al (2013) noted that in recent years there seems to be an unbounded interest concerning MSCs. This is mainly attributed to their exciting characteristics including long-term ex-vivo proliferation, multi-lineage potential and immunomodulatory properties. In this regard MSCs emerge as attractive candidates for various therapeutic applications. Mesenchymal stem cells were originally isolated from the BM and this population is still considered as the gold standard for MSC applications. However, the BM has several limitations as source of MSCs, including MSC low frequency in this compartment, the painful isolation procedure and the decline in MSC characteristics with donor's age. Thus, there is accumulating interest in identifying alternative sources for MSCs. To this end MSCs obtained from the Wharton's Jelly (WJ) of umbilical cords (UC) have gained much attention over the last years since they can be easily isolated, without any ethical concerns, from a tissue which is discarded after birth. Furthermore WJ-derived MSCs represent a more primitive population than their adult counterparts, opening new perspectives for cell-based therapies. The authors provided an overview of the biology of WJ-derived UC-MSCs; and discussed the potential application of WJ-derived UC-MSCs for the treatment of cancer and immune mediated disorders, such as GVHD and SLE.

Anti-Phospholipid Syndrome

Leone and colleagues (2017) noted that HSCT has been proposed as a therapeutic option for patients with SLE refractory to standard therapy. This therapeutic approach has been applied to other severe autoimmune diseases refractory to standard therapy with promising results. These investigators systematically reviewed the literature and analyzed the available evidence on HSCT therapy in patients with SLE and anti-phospholipid syndrome (APS), with a focus on effectiveness and occurrence of AEs. A detailed literature search, applied to Ovid Medline, In-Process and Other Non-Indexed Citation and Ovid Medline 1986 to 2014, has been developed a priori to identify articles that reported findings from clinical and laboratory studies that investigated the effect of HSCT in patients with SLE. A total of 25 studies met all inclusion criteria, including a total of 279 SLE patients; of those, 54 patients also fulfilled the classification criteria of APS. The majority of the studies reported an improvement after HSCT in terms of diseases activity control (assessed with SLE disease activity index [SLEDAI], or time-free from diseases) or OS. However, 1 study reported no net benefit of HSCT when compared to immunosuppression alone. One retrospective study reported an OS at 5 years of 81 % in 28 SLE patients. Of note, 5 cases (9.3 %) of aPL negativization were reported after HSCT in the APS patients. When combining these studies and analyzing these patients with APS, 32 out of 44 (73 %) were able to discontinue anti-coagulation after HSCT. These results also demonstrated a total of 86 infections in the pool of patients (30.8 %), 3 of which resulted in the death of the patient (1.3 %). These researchers observed an annual incidence of infection of 11.9 % with a mean follow-up of 36.2 months. The authors concluded that preliminary results of HSCT as a therapeutic option for SLE appeared promising. They stated that further studies are needed to evaluate the safety of the procedure for both the occurrence of secondary autoimmune disease and the rate of infection. However, the rate of AEs confines this option to very selected cases of SLE patients resistant or refractory to standard approaches.

Autoimmune Diseases-Induced Cirrhosis

Liang and colleagues (2017) stated that there has been great interest in recent years to take advantage of MSCs to treat end-stage liver disease. These researchers evaluated clinical therapeutic effects of allogeneic MSC transplantation in liver cirrhosis caused by autoimmune diseases. The enrolled patients with liver cirrhosis were assigned to receive allogeneic MSC infusions through a peripheral vein. The primary objective of this study was to assess the safety and effectiveness of MSCT in patients with autoimmune diseases-induced cirrhosis; secondary end-points were to assess changes in the Models of End Stage Liver Disease (MELD) scores and liver functions after the transplantation. A total of 26 patients were enrolled. Of

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

these, 23 patients received umbilical cord MSCT, 2 received cord blood MSCT, and 1 received bone marrow MSCT; 3 patents died of the complications caused by cirrhosis and 2 patients received liver transplantation after MSCT; 4 patients were lost to follow-up. The mean of alanine transaminase values decreased 6 months, 1 and 2 years after the transplantation, but there were no statistical significance. The mean value of total bilirubin decreased at 6 months and 1 year follow-up. Average serum albumin levels improved at 6 months, 1 and 2 years follow-up. The mean value at 2 years increased significantly compared with the baseline value. A lowering of prothrombin time was seen at 6 months after MSCT; MELD score improved at 6 months, 1 and 2 years of follow-up. No serious AEs were observed during or 24 hours after infusions of MSCs in any of the 26 patients with liver cirrhosis. The authors concluded that based on this clinical trial, allogeneic MSCT through the peripheral vein probably is safe and seemingly has beneficial effect in patients with liver cirrhosis. Thus, they stated that allogeneic MSCT is a potential option for the treatment of liver cirrhosis caused by autoimmune diseases; further studies with higher numbers of patients are needed to better clarify the impact and mechanisms of MSCT in liver cirrhosis.

Inflammatory Bowel Disease

Bakhtiar and associates (2017) stated that inflammatory bowel disease (IBD) in young children can be a clinical manifestation of various primary immunodeficiency syndromes. Poor clinical outcome is associated with poor quality of life and high morbidity from the complications of prolonged immunosuppressive treatment and malabsorption. In 2012, mutations in the lipopolysaccharide-responsive beige-like anchor (LRBA) gene were identified as the cause of an autoimmunity and immunodeficiency syndrome. Since then, several LRBA-deficient patients have been reported with a broad spectrum of clinical manifestations without reliable predictive prognostic markers. Allogeneic HSCT (allo-HSCT) has been performed in a few severely affected patients with complete or partial response. These researchers presented a detailed course of the disease and the transplantation procedure used in a LRBA-deficient patient suffering primarily from infantile IBD with immune enteropathy since the age of 6 weeks, and progressive autoimmunity with major complications following long-term immunosuppressive treatment. At 12 years of age, allo-HSCT using bone marrow of a fully matched sibling donor -- a healthy heterozygous LRBA mutant carrier -- was performed after conditioning with a reduced-intensity regimen. During the 6-year follow-up, these investigators observed a complete remission of enteropathy, autoimmunity, and skin vitiligo, with complete donor chimerism. The genetic diagnosis of LRBA deficiency was made after allo-HSCT by detection of 2 compound heterozygous mutations, using targeted sequencing of DNA samples extracted from peripheral blood before the transplantation. The authors concluded that due to the lacking genotype-phenotype correlation in LRBA-deficient patients, further data are needed to establish predictive and prognostic biomarkers that would allow the reliable identification of candidates for early allo-HSCT. they stated that further studies on transplantation in LRBA deficiency are needed to evaluate the optimal conditioning regimen and determine the necessary level of chimerism for disease remission.

Pemphigus

Wang and co-workers (2017) stated that pemphigus is a rare and fatal autoimmune disease for which the therapeutic options are limited. These researchers evaluated the effectiveness of autologous peripheral HSCT (APHSCT) for pemphigus. They conducted APHSCT for 12 pemphigus patients (7 males and 5 females, mean age of 23.8 years) with life-threatening complications or who responded poorly to conventional therapy. Peripheral blood stem cells were mobilized with cyclophosphamide, G-CSF, and rituximab, and purified autologous CD34+ stem cells were infused; OS, PFS, and AEs were recorded. With a mean follow-up period of 80.3 months, OS and complete clinical remission rates were 92 % (11/12) and 75 % (9/12), respectively; AEs included pyrexia, allergy, infection, and elevation of enzymes. Only 1 patient died of severe sepsis and multiple organ failure 2 months after APHSCT. The authors concluded that APHSCT is a promising therapeutic option for pemphigus.

Lupus Nephritis

Sattwika and associates (2018) stated that lupus nephritis (LN), a common manifestation of SLE, accounts for significant morbidity and mortality in SLE patients. Since the available standard therapies and biologic

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

agents for LN are yet to achieve the desired response and have considerable secondary effects, stem cell therapy has now emerged as a new approach. This therapy involves the transplantation of hematopoietic stem cells (HSCs) and MSCs. This review highlighted the progress of stem cell therapy for LN, along with the challenges encountered and the future direction of this approach.

Barbado and colleagues (2018) stated that animal and human studies have suggested the potential of MSCs for the treatment of SLE. These investigators presented the findings of compassionate MSC treatments for 3 SLE patients to provide the proof-of-concept for a RCT. Three patients of different ethnicities who suffered from chronic SLE, and who presented with class IV active proliferative nephritis confirmed by biopsy, were treated with allogeneic MSCs from healthy donors; 9 x 10(6) cells were infused intravenously into each patient during high and very high activity disease flare-ups and follow-up was continued for 9 months. Multi-organic affectation was quantified by the SLEDAI, and indicators of LN activity, such as proteinuria, as well as lymphocyte and monocyte antigens and anti-HLA antibodies were measured at 1, 3, 6, and 9 months after treatment. Proteinuria levels improved dramatically during the 1st month after treatment and the ameliorations were sustained throughout the follow-up period; SLEDAI scores revealed early, durable, and substantial remissions that were complete for 2 patients and partial for the 3rd patient and that permitted medication doses to be reduced 50 to 90 %. The authors concluded that these favorable outcomes supported completion of the randomized and controlled MSC trial for SLE.

Furthermore, an UpToDate review on “Treatment and prognosis of diffuse or focal proliferative lupus nephritis” (Falk et al, 2019) does not mention stem cell transplantation as a therapeutic option.

Autologous Mesenchymal Stem Cells for the Treatment of Multiple System Atrophy

Lee and associates (2012) stated that neuroprotective or regenerative strategies are invaluable in multiple system atrophy (MSA) due to its rapid progression with fatal prognosis. These researchers examined the efficacy of autologous MSC in patients with MSA-cerebellar type (MSA-C). A total of 33 patients with probable MSA-C and baseline Unified MSA rating scale (UMSARS) scores ranging from 30 to 50 were randomly assigned to receive MSC (4 × 10(7) /injection) via intra-arterial (IA) and intravenous (IV) routes or placebo. The primary outcome was change in the total UMSARS scores from baseline throughout a 360-day follow-up period between groups. Secondary outcomes were changes in the UMSARS part II scores, cerebral glucose metabolism, gray matter density, and cognitive performance over a 360-day period. The mixed model analysis of neurological deficits revealed a significant interaction effect between treatment group and time, suggesting that the MSC group had a smaller increase in total and part II UMSARS scores compared with the placebo group ($p = 0.047$ and $p = 0.008$, respectively). Cerebral glucose metabolism and gray matter density at 360 days relative to the baseline were more extensively decreased in the cerebellum and the cerebral cortical areas, along with greater deterioration of frontal cognition in the placebo group compared with the MSC group. These investigators found no serious AEs that were directly related to MSC treatment. However, IA infusion resulted in small ischemic lesions on MRI. The authors concluded that MSC therapy could delay the progression of neurological deficits in patients with MSA-C, suggesting the potential of MSC therapy as a treatment candidate of MSA.

Palma and Kaufmann (2015) noted that MSA is a sporadic, adult onset, relentlessly progressive neurodegenerative disease characterized by autonomic abnormalities associated with parkinsonism, cerebellar dysfunction, pyramidal signs, or combinations thereof. Treatments that can halt or reverse the progression of MSA have not yet been identified. MSA is neuropathologically defined by the presence ofα-synuclein-containing inclusions, particularly in the cytoplasm of oligodendrocytes (glial cytoplasmic inclusions, GCIs), which are associated with neurodegeneration. The mechanisms by which oligodendrocyticα-synuclein inclusions cause neuronal death in MSA are not completely understood. The MSA neurodegenerative process likely comprises cell-to-cell transmission ofα-synuclein in a prion-like manner,α-synuclein aggregation, increased oxidative stress, abnormal expression of tubulin proteins, decreased expression of neurotrophic factors, excitotoxicity and microglial activation, and neuroinflammation. In an attempt to block each of these pathogenic mechanisms, several pharmacologic approaches have been tried and shown to exert neuroprotective effects in transgenic mouse or cellular models of MSA. These include sertraline, paroxetine, and lithium, which hamper arrival ofα-synuclein to oligodendroglia; rifampicin, lithium, and non-steroidal anti-inflammatory drugs (NSAIDs), which inhibitα-

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

synuclein aggregation in oligodendrocytes; riluzole, rasagiline, fluoxetine and MSC, which exert neuroprotective actions; and minocycline and intravenous immunoglobulins (IVIGs), which reduce neuroinflammation and microglial activation.

In a phase-I/II clinical trial, Singer and colleagues (2019) examined intrathecal (IT) administration of MSCs as therapeutic approach to MSA. Utilizing a dose-escalation design, these researchers delivered between 10 and 200 million adipose-derived autologous MSCs intrathecally to patients with early MSA. Patients were closely followed with clinical, laboratory, and imaging surveillance. Primary end-points were frequency and type of AEs; key secondary end-point was the rate of disease progression assessed by the UMSARS. A total of 24 patients received treatment. There were no attributable serious AEs, and injections were generally well-tolerated. At the highest dose tier, 3 of 4 patients developed low back/posterior leg pain, associated with thickening/enhancement of lumbar nerve roots. Although there were no associated neurologic deficits, these researchers decided that dose-limiting toxicity (DLT) was reached. A total of 6 of 12 patients in the medium dose tier developed similar, but milder and transient discomfort. Rate of progression (UMSARS total) was markedly lower compared to a matched historical control group (0.40 ± 0.59 versus 1.44 ± 1.42 points/month, $p = 0.004$) with an apparent dose-dependent effect. The authors concluded that IT MSC administration in MSA was safe and well-tolerated but could be associated with a painful implantation response at high doses; compelling dose-dependent efficacy signals are the basis for a planned placebo-controlled trial. This phase-I/II clinical trial provided Class IV evidence that for patients with early MSA, IT MSC administration was safe, may result in a painful implantation response at high doses, and was associated with dose-dependent efficacy signals.

Furthermore, an UpToDate review on “Multiple system atrophy: Prognosis and treatment” (Factor and Esper, 2020) states that “Investigational Therapies -- There is currently no effective disease-modifying or neuroprotective treatment for MSA. Several therapies have failed in phase III trials. Some strategies currently under investigation in clinical trials include reducing alpha-synuclein aggregation using serotonin specific reuptake inhibitors such as sertraline, inhibiting neuroinflammatory response with intravenous immunoglobulins and other agents, and infusing autologous mesenchymal stem cells, which theoretically may have a neuroprotective effect. Emerging studies include a phase I study of active vaccination against an epitope similar to alpha-synuclein”.

Autologous Umbilical Cord Blood Stem Cells for the Treatment of Autism

Dawson et al (2017) stated that despite advances in early diagnosis and behavioral therapies, more effective treatments for children with autism spectrum disorder (ASD) are needed. These researchers hypothesized that umbilical cord blood (UCB)-derived cell therapies may have potential in alleviating ASD symptoms by modulating inflammatory processes in the brain. Accordingly, they carried out an open-label, phase-I clinical trial to examine the safety and feasibility of a single intravenous (IV) infusion of autologous UCB, as well as sensitivity to change in several ASD assessment tools, to determine suitable endpoints for future trials. A total of 21 children, median age of 4.6 years (range of 2.26 to 5.97), with a confirmed diagnosis of ASD and a qualified banked autologous UCB unit, were enrolled. Children were examined with a battery of behavioral and functional tests immediately prior to UCB infusion (baseline) and 6 and 12 months later. Assessment of adverse events (AEs) across the 12-month period indicated that the treatment was safe and well-tolerated. Significant improvements in children's behavior were observed on parent-report measures of social communication skills and autism symptoms, clinician ratings of overall autism symptom severity and degree of improvement, standardized measures of expressive vocabulary, and objective eye-tracking measures of children's attention to social stimuli, indicating that these measures may be useful endpoints in future studies. Behavioral improvements were observed during the first 6 months after infusion and were greater in children with higher baseline non-verbal intelligence quotients. The authors concluded that these findings will serve as the basis for future studies to determine the efficacy of UCB infusions in children with ASD.

In an exploratory study, Chez et al (2018) examined the safety and clinical effects of autologous UCB (AUCB) infusion in children with idiopathic ASD. A total of 29 children 2 to 6 years of age with a confirmed diagnosis of ASD participated in this randomized, blinded, placebo-controlled, cross-over trial. Subjects were randomized to receive AUCB or placebo, evaluated at baseline, 12, and 24 weeks, received the

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

opposite infusion, then re-evaluated at the same time-points. Evaluations included assessments of safety, Expressive One Word Picture Vocabulary Test, 4th edition, Receptive One Word Picture Vocabulary Test, 4th edition, Clinical Global Impression, Stanford-Binet Fluid Reasoning and Knowledge, and the Vineland Adaptive Behavior and Socialization Subscales. Generalized linear models were used to examine the effects of the response variables at the 12- and 24-week time periods under each condition (AUCB, placebo). There were no serious AEs. There were trends toward improvement, particularly in socialization, but there were no statistically significant differences for any endpoints. The authors concluded that the findings of this study suggested that AUCB infusions were safe for children with ASD. Moreover, these researchers stated that tightly controlled trials are needed to further progress the study of AUCB for autism.

Carpenter et al (2019) noted that ASD is a heterogeneous neurodevelopmental disorder characterized by social communication deficits and the presence of restricted interests and repetitive behaviors. These investigators have previously reported significant improvements in behavior, including increased social functioning, improved communication abilities, and decreased clinical symptoms in children with ASD, following treatment with a single infusion of AUCB in an open-label, phase-I clinical trial. In the current study, these researchers examined if these improvements were associated with concurrent changes in brain structural connectivity. A total of 25 2- to 6-year-old children with ASD participated in this study. Clinical outcome measures included the Vineland Adaptive Behavior Scales-II Socialization Subscale, Expressive One-Word Picture Vocabulary Test-4, and the Clinical Global Impression-Improvement Scale. Structural connectivity was measured at baseline and at 6 months in a subset of 19 children with 25-direction diffusion tensor imaging and deterministic tractography. Behavioral improvements were associated with increased white matter connectivity in frontal, temporal, and subcortical regions (hippocampus and basal ganglia) that have been previously shown to show anatomical, connectivity, and functional abnormalities in ASD. The authors concluded that the findings of this study suggested that improvements in social communication skills and a reduction in symptoms in children with ASD following treatment with AUCB infusion were associated with increased structural connectivity in brain networks supporting social, communication, and language abilities.

The authors stated that this study had several drawbacks. First, because this was an exploratory analysis, these researchers restricted their analyses to brain networks that have previously been implicated in autism. Furthermore, they only reported those findings where there was increased connectivity that correlated with at least 2 of their outcome measures. Previous work examining brain changes in response to treatment with UCB in children with cerebral palsy (CP) demonstrated strong correlations between increased whole brain white matter connectivity and improved functional abilities in children with CP. Although these investigators did not find evidence of such a pattern in the current data, it is possible that other regions that these researchers did not examine also showed changes in mutual connectivity. These investigators stated that future work examining this may provide an even greater picture of the brain networks associated with behavioral improvement following treatment.

In a prospective, randomized, placebo-controlled, double-blind, phase-II clinical trial, Dawson et al (2020) examined if UCB infusion is safe and associated with improved social and communication abilities in children with ASD. This study included 180 children with ASD, aged 2 to 7 years, who received a single IV autologous (n = 56) or allogeneic (n = 63) UCB infusion versus placebo (n = 61) and were evaluated at 6 months post-infusion. UCB infusion was safe and well-tolerated. Analysis of the entire sample showed no evidence that CB was associated with improvements in the primary outcome, social communication (Vineland Adaptive Behavior Scales-3 [VABS-3] Socialization Domain), or the secondary outcomes, autism symptoms (Pervasive Developmental Disorder Behavior Inventory) and vocabulary (Expressive One-Word Picture Vocabulary Test). There was also no overall evidence of differential effects by type of UCB infused. In a sub-analysis of children without intellectual disability (ID), allogeneic, but not autologous, UCB was associated with improvement in a larger percentage of children on the clinician-rated Clinical Global Impression-Improvement scale; however, the odds ratio (OR) for improvement was not significant. Children without ID treated with UCB showed significant improvements in communication skills (VABS-3 Communication Domain), and exploratory measures including attention to toys and sustained attention (eye-tracking) and increased alpha and beta electroencephalographic power. The authors concluded that a single IV infusion of UCB was not associated with improved socialization skills or reduced autism symptoms.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

These investigators stated that more research is needed to examine if UCB infusion is an effective treatment for some children with ASD.

Furthermore, an UpToDate review on "Autism spectrum disorder in children and adolescents: Overview of management" (Weissman, 2021) does not mention umbilical cord blood as a management / therapeutic option.

Autologous Hematopoietic Stem Cell Transplantation for the Treatment of Stiff-Person Spectrum Disorder

In a prospective, open-label, pilot study, Burt and colleagues (2021) examined the safety and efficacy autologous non-myeloablative HSCT in the treatment of stiff-person spectrum disorder (SPSD). A total of 23 subjects were treated in this clinical trial. After stem cell mobilization with cyclophosphamide (2 g/m2) and filgrastim (5-10 µg/kg/d), subjects were treated with cyclophosphamide (200 mg/kg) divided as 50 mg/kg IV on day -5 to day -2; rabbit anti-thymocyte globulin (thymoglobulin) given intravenously at 0.5 mg/kg on day -5, 1 mg/kg on days -4 and -3, and 1.5 mg/kg on days -2, and -1 (total dose 5.5 mg/kg); and rituximab 500 mg IV on days -6 and +1. Unselected peripheral blood stem cells were infused on day 0. Safety was evaluated by survival and National Cancer Institute (NCI) common toxicity criteria for AEs during HSCT. Outcome was examined by greater than or equal to 50 % decrease or discontinuation of anti-spasmodic drugs and by QOL instruments. There was no treatment-related mortality; 1 subject died 1 year after transplantation of disease progression. Of the 74 % of subjects who responded, 47 % have stayed in remission for a mean of 3.5 years; 26 % did not respond. Compared to non-responders, responders were more likely to have pre-transplantation intermittent muscle spasms (16 of 17 versus 0 of 6), normal reflexes (12 of 17 versus 0 of 6), and positive cerebrospinal fluid (CSF) anti-glutamic acid decarboxylase serology (12 of 14 versus 2 of 6). Compared to responders, non-responders were more likely to have lead pipe rigidity (4 of 6 versus 0 of 17) and electromyography (EMG)-documented simultaneous contraction of agonist/antagonist limb muscles (4 of 6 versus 1 of 17). Pre-HSCT use of prescription serotonin selective receptor inhibitor (SSRI) or serotonin and norepinephrine reuptake inhibitor (SNRI) was more common in those who relapsed or never responded (9 of 12) compared to those responders who never relapsed (0 of 11). The authors concluded that in this cohort, HSCT was safe, but the beneficial effect of HSCT was variable and confined predominately to subjects with episodic spasms and normal tendon reflexes without simultaneous co-contraction of limb agonist/antagonist muscles who were not taking SSRI or SNRI anti-depressants. This study provided Class IV evidence that, for a subset of people with SPSD, autologous nonmyeloablative HSCT improved outcomes. These researchers stated that the drawbacks of this trial included the retrospective data mining to identify responders, the single-center nature, and the small number of subjects (n = 23).

Pluripotent Stem Cell-Based Retinal Organoids for Restoration of Blindness

M'Barek and associates (2019) noted that in developed countries, blindness and visual impairment are caused mainly by diseases affecting the retina. These retinal degenerative diseases, including age-related macular dystrophy (ARMD) and inherited retinal diseases (IRDs) such as retinitis pigmentosa (RP), are the predominant causes of human blindness worldwide and are responsible for more than 1.5 million cases in France and more than 30 million cases worldwide. Global prevalence and disease burden projections for next 20 years are alarming, and strongly argue toward designing innovative eye-care strategies. Currently, despite the scientific advances achieved in the past years, there is no cure for such diseases, making retinal degenerative diseases an unmet medical need. The majority of IRDs genes codes for proteins acting directly in photoreceptors (PRs); yet, a few of them are expressed in the retinal pigment epithelium (RPE), the supporting tissue necessary for proper functioning of the PRs. Among retinal degenerative diseases, impairment of some RPE genes engenders a spectrum of conditions ranging from stationary visual defects to very severe forms of retinal dystrophies in which the RPE dysfunction leads to PRs cell death and consecutive irreversible vision loss. The accessibility of the eye and the immune privilege of the retina, together with the availability of non-invasive imaging technologies, make such inherited retinal dystrophies a particularly attractive disease model for innovative cell therapy approaches to replace, regenerate, and/or repair the injured RPE tissue. Proof-of-concept studies in animal models have demonstrated the safety and

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

efficacy of the engraftment of therapeutic cells either to support RPE cell functions or to provide a trophic support to PRs. These different approaches are now in the pipeline of drug development with objective to provide first cell-based treatments by 2020.

Singh and colleagues (2021) stated that blindness is caused by a number of conditions with high prevalence such as glaucoma, ARMD, and diabetic retinopathy (DN). Furthermore, a large number of orphan-inherited (mostly monogenic) conditions, such as RP and Leber congenital amaurosis (LCA), add to the overall number of patients with blinding retinal degenerative diseases. Blindness caused by deterioration and loss of retina is so far incurable. Modern biomedical research leveraging molecular and regenerative medicine approaches had a number of ground-breaking discoveries and proof-of-principle treatments of blindness in animals; however, these methods are slow to be standardized and commercialized as therapies to benefit individuals losing their eyesight due to retinal degenerative conditions. These investigators outlined major regenerative medicine approaches, which are emerging as promising for preserving or/and restoring vision. They discussed the potential of each of these approaches to reach commercialization step and be converted to treatments, which could at least ameliorate blindness caused by retinal cell death. One of the promising approaches in this field is human pluripotent stem cells (hPSCs)-based organoid technology. These researchers stated that retinal organoid technology and the ability to derive human retinal tissue in a dish from hPSCs brought a lot of promise to regenerative medicine focused on restoring vision. Retinal organoids undergo self-formation when hPSCs are induced to differentiate toward neural and retinal lineage. Large-scale biomanufacturing of retinal organoids can be developed at a relatively inexpensive cost. Young retinal organoids (8 to 12 weeks following induction) carry all types of retinal cells and layers, typical for developing human fetal retina. Just like in human fetal retinal tissue, retinal organoids carry a developing layer of PRs, also 2nd order neurons/progenitors of retinal interneurons, and retinal ganglion cells (RGCs). The only exception appeared to be RPE, which is always present in developing human fetal retina, but may be either completely absent or present as patches; and still does not cover the whole neural retina. Compared to human fetal retina, hPSC-derived retinal organoids carry no ethical baggage associated with clinical application of human fetal retina or retinal cells; and provide re-plenishable source of human retinal cells and retinal tissue for retinal therapies aimed at rebuilding degenerated retina and slowing down vision loss. The authors concluded that retinal organoid-derived therapies could potentially address both early (epiretinal graft neurotrophic effect) and late stage (cell replacement) diseases. These researchers stated that as with other promising approaches, further pre-clinical work and refinement of retinal organoid technologies are needed to develop them toward clinical trials and commercialization.

Treatment of Ankylosing Spondylitis and Sjogren's Syndrome with Mesenchymal Stem Cells

Chihaby et al (2021) stated that MSCs are ubiquitous in the human body. Mesenchymal stem cells were initially isolated from bone marrow and later from other organs such as fatty tissues, umbilical cords, and gingiva. Their secretory capacities give them interesting immunomodulatory properties in cell therapy. Some studies have examined the use of MSCs to treat Sjogren's syndrome (SS), a chronic inflammatory autoimmune disease that mainly affects exocrine glands, including salivary and lacrimal glands, although current treatments are only palliative. In a systematic review, these investigators examined the current data regarding the use of MSCs in SS. Reports showed improvements in salivary secretions and a decrease in lymphocytic infiltration in salivary glands in patients and mice with SS after intravenous or infra-peritoneal injections of MSCs. MSC injections led to a decrease in inflammatory cytokines and an increase in anti-inflammatory cytokines; however, the intrinsic mechanism of action of these MSCs currently remains unknown. The authors concluded that additional clinical studies are needed to provide more powerful scientific evidence, such as long-term effects on clinical outcomes and safety, the duration of therapy, the optimal timing, and the dosages for treating SS.

In a systematic review and meta-analysis, Zeng et al (2022) examined the safety and effectiveness of MSC transplantation in the treatment of autoimmune diseases. The Chinese and English databases were searched for clinical research on the treatment of autoimmune diseases with MSCs. The search time range was from a self-built database to October 1, 2021. Two reviewers independently screened the literature according to the inclusion and exclusion criteria, extracted data, and evaluated the bias of the included studies. RevMan 5.3

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

analysis software was used for meta-analysis. A total of 18 RCTs involving 5 autoimmune diseases were included. The 5 autoimmune disease were ankylosing spondylitis (AS), IBD, MS, RA, and SLE. For RA, the current RCTs still believe that stem cell transplantation may reduce disease activity, improve the clinical symptoms (such as DAS28), and the percentage of CD4+CD 25+Foxp3+Tregs in the response group increased and the percentage of CD4+IL-17A+Th17 cells decreased. The total clinical effective rate of RA was 54 %. For SLE, the results showed that MSC transplantation may improve SLEDAI [-2.18 (-3.62 to -0.75), p = 0.003], urine protein [-0.93 (-1.04 to -0.81), p < 0.00001], and complement C3 [0.31 (0.19 to 0.42), p < 0.00001]. For IBD, the results showed that MSC transplantation may improve clinical effectiveness [2.50 (1.07 to 5.84), p = 0.03]. For AS, MSC treatment for 6 months may increase the total effective rate; reduce erythrocyte sedimentation rate (ESR), intercellular adhesion molecules, and serum tumor necrosis factor-alpha (TNF-α); and improve pain and activity. For MS, the current research results are still controversial, so more RCTs are needed to amend or confirm the conclusions. No obvious AEs of MSC transplantation were found in all RCTs. The authors concluded that MSCs exhibited a certain effect on different autoimmune diseases; however, more RCTs are needed to further modify or confirm the conclusion.

Cogan Syndrome

An UpToDate review on “Cogan syndrome” (St Clair and McCallum, 2023) does not mention autologous stem cell transplant as a management / therapeutic option.

Inflammatory Myopathies, Myasthenia Gravis, and Stiff-Person Syndrome

Jaime-Perez et al (2022) noted that CIDP, Stiff-person syndrome (SPS), neuromyelitis optica spectrum disorders (NMOSD) and severe refractory myasthenia gravis (MG) are immune-mediated neurological diseases that severely affect patients' functionality and QOL, with a considerable percentage undergoing relapse or not responding to conventional therapeutic options; AHSCT has emerged as a potential 2nd-line treatment alternative. These investigators carried out a literature review in PubMed/Medline, Embase, Web of Science and Cochrane Library from inception to September 2021 of reported cases and studies of CIDP, SPS, NMOSD and MG that underwent HSCT as a therapeutic option. A total of 173 patients who underwent HSCT were found, including 32 patients described in case reports and 60 in a phase-II clinical trial with CIDP, 29 patients with SPS, 42 patients with NMOSD and 10 patients with refractory MG. Complete remission (CR) was reported in 68/92 patients with CIDP, 13/29 with SPS and 10/10 with MG. From the NMOSD cases, 24/42 were relapse-free at last follow-up, with 13/33 having negative anti-AQ4 antibodies after HSCT. From all the included studies, only 8/173 patients received an allo-HSCT, 4/8 after a failed AHSCT. All showed clinical improvement and disease remission. The authors concluded that HSCT has the potential to induce long-term remission in patients with CIDP, NMOSD, SPS or MG with adequate safety and tolerability. Collaboration between centers is needed to implement larger, homogeneous prospective studies, focusing on immunological correlates of favorable long-term response.

Mariottini et al (2023) stated that over the past 20 years, HSCT has been examined as a potential therapeutic strategy for autoimmune diseases refractory to conventional treatments, including neurological disorders. Although both AHSCT and allo-HSCT were investigated, AHSCT was preferentially developed due to a more favorable safety profile compared to allo-HSCT. In this study, these investigators discussed the rationale for the use of HSCT in neurological diseases and the experimental models that prompted its clinical application. Case series and prospective studies examining the use of HSCT in autoimmune diseases other than MS were discussed, covering both frequent and rare neurological disorders such as inflammatory myopathies (including dermatomyositis, polymyositis, and sporadic late-onset nemaline myopathy), MG, and SPS. Finally, an updated summary of ongoing and future studies focusing on this issue was provided.

Beland et al (2025) stated that patients with refractory MG have few therapeutic options. Autologous HSCT has been used for the treatment of patients with immune diseases; however, its use in the treatment of MG is not broadly considered. In a retrospective study, these researchers reported on the safety and effectiveness of AHSCT in the treatment of refractory MG. A total of 21 patients who underwent AHSCT for MG were examined. All subjects had severe MG refractory to multiple therapies. Stem cells were mobilized with cyclophosphamide and G-CSF. The grafts were depleted of immune cells by selecting CD34+ cells. AHSCT

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

conditioning consisted of high-dose cytoreductive therapy and anti-thymocyte globulin (ATG). The primary effectiveness outcome was achieving clinically stable remission or minimal manifestations without treatment and remaining as such until most recent follow-up. The median time from MG diagnosis to AHSCT was 4.0 years. The primary outcome was reached in 16 of 18 evaluable patients (89 %) at a median of 1.7 years and maintained with a median follow-up of 6.7 years (range of 1.0 to 21.9 years). Three patients were not evaluable for the primary outcome: 1 due to confounding illness and 2 died within 12 months of transplant. The transplant-related mortality at 100 days was 9.5 %; and 2 late deaths occurred, with uncertain relation to the AHSCT. The authors concluded that after AHSCT for refractory MG, most patients achieved sustained disease remission; however, AHSCT-related mortality in medically complex MG patients may be high. These researchers stated that prospective, high-quality, phase-II clinical studies are needed to examine the safety and effectiveness of AHSCT in the treatment of refractory MG.

The authors stated that this study had several drawbacks. First, the retrospective nature of the study, associated with a lack of prospective evaluation of autoantibody levels after HSCT. Second, patients included in this trial were not treated with novel MG therapies that are now often used for severe or refractory MG (rMG), except for 1 subject who failed eculizumab. Eculizumab is a humanized monoclonal antibody that targets the terminal complement protein, C5. The Myasthenia Gravis Foundation of America Post-intervention Status (MGFA-PIS) reported in the interim analysis of the REGAIN Trial suggested that 56.0 % achieved minimal manifestations (MM) or pharmacologic remission (PR) and 74.1 % had improvement in their disease status. MGFA-PIS was not reported in the extension trial. Although it would not be appropriate to make direct comparisons between the small cohort and this trial, it was apparent that none of the patients in the REGAIN Trial achieved disease remission or were able to remain in treatment-free remission, which was the case for the majority of the patients in this study. Whereas with HSCT , these researchers expected near normal immune reconstitution following auto-HCT, ongoing therapy with eculizumab would result in a persistent immune defect against encapsulated bacteria, among other pathogens. Efgartigimod is a human IgG antibody that functions to reduce the recycling of auto-inflammatory IgG molecules by inhibiting the binding of IgG to the neonatal Fc receptor. In the ADAPT Trial, not all patients were refractory to treatment; 23 out of 167 (13.8 %) had not received either steroid or non-steroid immunosuppression, 40 out of 167 (24.0 %) did not receive steroid, and 65 out of 167 (38.9 %) did not receive any non-steroid immunosuppression. Although efgartigimod is effective at improving MG activities of daily living (MG-ADL) MG-ADL and quantitative MG (QMG) scores in patients with severe MG, it is unknown whether similar effectiveness would be observed in patients with refractory disease, like those included in the cohort in this trial. Although no studies of QOL are available, QOL after autologous HCT for malignancy appeared to return to population norm by 1-year post-transplant. Third, the cohort in this study was heterogenous in the treatments that they received before transplantation. This may have implications on their disease status before HSCT, especially as many received their transplantation before the availability of rituximab in the authors' region (although 9 out of 20 patients failed rituximab) and none received efgartigimod. However, the heterogeneity reflected the real-world experience in treating MG and the improvement in disease control resulting in sustained clinically stable remission (CSR)/MM-0 in of the cohort despite the heterogeneity spoke to the effectiveness of HSCT in rMG. The HSCT conditioning regimens were not completely uniform among the cohort in this study and the low numbers of alternative conditioning regimens made it difficult to compare one to another. Fourth, the incomplete information for some patients and, although this was the largest cohort of MG patient treated with HSCT, it was still a small sample size that may have implications on the generalizability of the data. Fifth, the diagnosis of MG was made based on a combination of clinical, serological, and/or electrophysiological findings in patients who came from multiple provinces, multiple healthcare systems, over multiple decades before electronic medical records were available. These investigators did not have results for electrophysiologic testing around the time of diagnosis for one of the patients who was sero-negative; however, their clinical presentation was consistent with MG.

Neuromyelitis Optica Spectrum Disorder

Nabizadeh et al (2022) noted that therapeutic options for neuromyelitis optica spectrum disorder (NMOSD) are corticosteroids, immunosuppressive drugs, emerging monoclonal antibodies, rituximab, eculizumab, satralizumab, and inebilizumab. Due to disabling and deadly nature of NMOSD, there is a great motivation

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

among physicians for finding new therapies. Recently, several studies have been carried out on the therapeutic effects of AHSCT on NMOSD patients. Several databases including PubMed, Scopus, Web of Science, and Google scholar were searched for studies on AHSCT in patients with NMOSD. After screening titles and abstracts, and reviewing full texts, a total of 9 studies with 39 severe cases of NMOSD met the criteria of this systematic review and meta-analysis. The pooled standardized mean difference (SMD) for EDSS score before and after treatment was -0.81 (95 % CI: -1.07 to -0.15; Q = 1.99, p = 0.58, I2 = 0 %). Furthermore, the PFS and relapse-free survival (RFS) were 69 % and 53 %, respectively (PFS: 69 %, 95 % CI: 42 % to 96 %; Q = 8.63, p = 0.01, I2 = 73.07 %; RFS: 53 %, 95 % CI: 27 % to 79 %; Q = 12.33, p = 0.01, I2 = 71.87 %). In addition, there were 3 cases with secondary autoimmune diseases including myasthenia gravis, hyperthyroidism, and thyroiditis. The authors concluded that according to the findings of this study, AHSCT could be an alternative therapy for NMOSD in severe cases instead of conventional immunotherapies.. However, physicians should pay attention to its serious complications. The diversity of results from the published trials on the safety and effectiveness of AHSCT warrants further investigations on determining the ideal AHSCT conditioning and the characteristics of patients.

Allogeneic Hematopoietic Stem Cell Transplantation for Behcet's Disease with Myelodysplastic Syndrome

Lei et al (2024) stated that trisomy 8 is a frequent chromosomal abnormality that occurs among patients with myelodysplastic syndrome (MDS). This abnormality is more common among MDS patients with concurrent Behcet's disease (BD) than among patients with MDS alone. This co-morbidity is also associated with a higher prevalence of intestinal ulcers. In a single-case study, these researchers described a 45-year-old woman who was diagnosed with BD at age 35. Eight years later, her BD relapsed with trisomy 8-positive MDS. After the disease was controlled by azacitidine, the subject received allo-HSCT, which entailed the use of both peripheral blood stem cells and bone marrow stem cells. Although the subject experienced grave oral mucositis and chronic intestinal GVHD, remission of both MDS and BD was maintained. The authors concluded that the findings of this case-report highlighted the potential effectiveness of allo-HSCT in the treatment of patients with refractory autoimmune diseases complicated by MDS, especially patients with trisomy 8 abnormalities. In addition, these investigators discussed the immunological implications and suggested allo-HSCT as a viable curative option for patients with BD and MDS.

Allogeneic Hematopoietic Stem Cell Transplantation for Juvenile Idiopathic Arthritis and Associated Lung Disease

Matt et al (2025) noted that systemic juvenile idiopathic arthritis-related lung disease (sJIA-LD) is a severe complication in patients with treatment-refractory sJIA. In a retrospective study, these researchers examined the effect of allogeneic HSCT in a cohort of children with sJIA-LD. This trial was carried out in 9 hospitals across the U.S. and Europe in children with sJIA-LD who had received allogeneic HSCT. Patients' medical charts were reviewed and their data extracted using a standardized form. The outcomes assessed were allogeneic HSCT outcomes (e.g., engraftment and donor chimerism, and transplant-related complications), pulmonary outcomes (e.g., oxygen dependence, chest CT findings, and pulmonary function test results), and overall outcomes (e.g., death, CR, or partial response [PR]). A CR was defined as resolution of signs of sJIA without the need for systemic immunomodulatory therapy, in addition to discontinuation of supplemental oxygen. Between January 18, 2018, and October 17, 2022, a total of 13 patients with sJIA-LD, who were refractory to immunosuppressive treatment and who had received an average of 6 different treatment agents, underwent allogeneic HSCT – 10 (77 %) of 13 patients were female and 3 (23 %) were male. The median age at diagnosis of sJIA-LD was 4·8 years (IQR 2.9 to 14.8) and the median age at transplantation was 9.0 years (5.0 to 19.0). Pre-HSCT chest CT showed characteristic sJIA-LD. A total of 5 patients required supplemental oxygen before transplantation. Subjects received various reduced toxicity or intensity conditioning regimens. Grafts were from 10/10 HLA-matched (n = 6) or 9/10 HLA-mismatched (n = 5) unrelated donors, a 7/10 related donor (n = 1), and a matched sibling (n = 1). All patients engrafted – 1 patient had secondary graft failure and received a 2nd transplant from a different donor. Post-transplantation complications included acute GVHD (n = 5), bacteremia (n = 8), CMV re-activation (n = 6), and post-transplantation macrophage activation syndrome (n = 3). A total of 4 patients died – 2 from CMV pneumonitis, 1 rom intra-cranial hemorrhage, and 1 from progressive sJIA-LD. At a median follow-up of 16

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

months (IQR 6 to 24), all 9 surviving patients had a CR, with no active features of sJIA, no biological therapy or corticosteroid use, and no supplemental oxygen dependence. The authors concluded that allogeneic HSCT might be a valuable therapeutic option for patients with refractory sJIA and sJIA-LD and should be considered for children with worsening oxygen dependence or severe treatment-related morbidity.

An UpToDate review on "Systemic juvenile idiopathic arthritis: Treatment and prognosis" (Kimura and Schulert, 2025) list HSCT as one of the treatment strategies may be used in patients who fail the more standard therapies. However, HSCT was primarily used prior to the advent of effective biologic therapies. Given its high mortality rate, the use of HSCT is restricted to only the most severely affected children refractory to other, less risky therapies. Moreover, HSCT is not mentioned in the "Summary and Recommendations" section of this UTD review.

Autologous Hematopoietic Stem Cell Transplantation for Autoimmune Retinopathy

Wong et al (2025) noted that autoimmune retinopathy (AIR) is a rare, potentially blinding retinal disease that remains a challenging condition to manage when resistant to conventional immune-modulatory approaches. These investigators reported clinical and electrophysiological improvement of a 49-year-old woman who underwent AHSCT for thymoma-associated AIR after experiencing progressive disease despite receiving peri-ocular as well as systemic steroids, mycophenolate mofetil, baricitinib, tacrolimus, bortezomib, rituximab, plasmapheresis, and IVIG. The AHSCT had 2 stages. First -- peripheral blood stem cell harvest following mobilization with cyclophosphamide and G-CSF. Second --conditioning regimen with plasmapheresis, rituximab, cyclophosphamide, and ATG high-dose therapy, followed by autologous hematopoietic cell infusion of 5.74 million cells. Symptoms of photopsia rapidly abated after undergoing AHSCT, and objective evaluations of structure and function revealed improvement. At 22-month follow-up, the patient continued to show the durability of the clinical response. The authors concluded that the findings of this single-case study suggested that in judiciously selected patients, AHSCT may provide a rescue option for refractory AIR. Moreover, these researchers stated that further investigations are needed to confirm AHSCT as a viable rescue option for the treatment of refractory AIR.

HLA-Haploidentical Stem Cell Transplantation for Chronic Granulomatous Disease

Riller et al (2025) stated that chronic granulomatous disease (CGD) is an inborn error of immunity characterized by defective NADPH oxidase function, resulting in impaired microbial killing, recurrent infections as well as granulomatous inflammation. Allogenic HSCT is a curative treatment for CGD, especially effective when a fully HLA-matched donor is available. However, the place of HLA-haploidentical HSCT remains less established. In a retrospective, multi-center study, these researchers examined outcomes of 64 CGD patients (53 males, 46 with X-linked CGD) who underwent a 1st HSCT with HLA-haploidentical family donors either with in-vitro TCRαβ/CD19 depletion or in-vivo depletion using post-transplant cyclophosphamide (PTCY). The mean age at transplant was 5.8 years (0 to 33 years). Patients exhibited a high disease burden before HSCT, with 45 % experiencing infections in the 6 months before HSCT and 67 % exhibiting inflammation. Outcomes in the entire cohort showed a 3-year OS, EFS, and GVHD grade III to IV-free EFS (GEFS) of 75.9 %, 70.2 %, and 56.1 %, respectively; and were not impacted by the type of depletion or age. The cumulative incidence of primary graft failure was 20.6 %. The cumulative incidence of grade II to IV acute GVHD was higher in the PTCY group ($p = 0.04$) whereas the cumulative incidence of GVHD grade III to IV was not. These findings indicated that HLA-haploidentical HSCT was a feasible transplant option for CGD patients lacking HLA-matched donors. Moreover, these researchers stated that further refinement of transplant protocols is needed to mitigate graft failure and acute GVHD, ultimately improving access and outcomes for this life-saving therapy.

Umbilical Cord Blood Transplantation in Children with Neuronal Ceroid Lipofuscinosis

Bauchat et al (2025) noted that ceroid lipofuscinosis neuronal (CLN) entails rare inherited neurodegenerative disorders that present in childhood with clinical features including epilepsy, psychomotor delay, progressive vision loss, as well as pre-mature death. Available evidence on the use of umbilical cord blood transplant

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

(UCBT) for these disorders is limited. In a retrospective, these researchers examined patients with CLN (CLN2, CLN3, and CLN5) who underwent UCBT from 2012 to 2020. All participants (n = 8) received SOC myeloablative conditioning; and 4 also enrolled in clinical trial NCT02254863 and received intra-thecal DUOC-01 cells post-transplant. Median age at UCBT was 5.9 years. All subjects achieved neutrophil engraftment with greater than 95 % donor chimerism at a median of 28.5 days; and sinusoidal obstructive syndrome was not observed. Severe acute GVHD occurred in 12.5 %; other complications included autoimmune hemolytic anemia (25 %) and viral reactivation/infection (62.5 %). No transplant-related mortality was observed. Two CLN2 patients died, 1 from progressive disease, and 1 from unknown cause at days +362 and +937, respectively. With median follow-up of 8 years, OS at 100 days and 24 months was 100 % and 88 %, respectively. A total of 3 of 4 CLN3 subjects stabilized Hamburg motor and language scores. The authors concluded that while UCBT appeared safe and feasible in these patients, given the variable expression and natural history, extended follow-up and further studies are needed to ascertain the potential impact of UCBT on clinical outcomes.

These investigators stated that while this trial was limited by sample size (n = 8) and the heterogeneity of NCLs, UCBT appeared a safe and feasible therapeutic option. No patient in this cohort died from transplant-related complications; GVHD occurred but was treatable, graft rejection was not observed, and all patients had full donor chimerism at last evaluation. Inferences regarding donor cell doses, risk of GVHD and other AEs such as hemolytic anemia were limited by the small cohort and heterogeneity. Nevertheless, in this cohort, outcomes appeared poorer in those with advanced stage disease, rapidly progressing disease, and/or CLN2 diagnosis. Later disease onset, slower progressing disease, and/or CLN3 may correlate with disease stabilization although this may coincide with the natural history of CLN3 patients. Reports on cellular, gene, and enzyme therapies for the treatment of NCLs are emerging. Intra-vitreal enzyme gene therapy is emerging as progressive vision loss is not typically improved by enzyme replacement. Systemic disease modification is needed with this adjunctive approach given the CNS pathologic contribution to functional vision.

References

The above policy is based on the following references:

1. Alaez C, Loyola M, Murguia A, et al. Hematopoietic stem cell transplantation (HSCT): An approach to autoimmunity. Autoimmun Rev. 2006;5(3):167-179.
2. Al-toma A, Nijeboer P, Bouma G, et al. Hematopoietic stem cell transplantation for non-malignant gastrointestinal diseases. World J Gastroenterol. 2014;20(46):17368-17375.
3. Appel SH, Armon C. Stem cells in amyotrophic lateral sclerosis: Ready for prime time? Neurology. 2016;87(4):348-349.
4. Atkins H. Hematopoietic SCT for the treatment of multiple sclerosis. Bone Marrow Transplant. 2010;45(12):1671-1681.
5. Atkins H. Stem cell transplantation to treat multiple sclerosis. JAMA. 2019 Jan 15;321(2):153-155.
6. Atkins HL, Bowman M, Allan D, et al. Immunoablation and autologous haemopoietic stem-cell transplantation for aggressive multiple sclerosis: A multicentre single-group phase 2 trial. Lancet. 2016;388(10044):576-585.
7. Bakhtiar S, Gamez-Diaz L, Jarisch A, et al. Treatment of infantile inflammatory bowel disease and autoimmunity by allogeneic stem cell transplantation in LPS-responsive beige-like anchor deficiency. Front Immunol. 2017;8:52.
8. Bakhuraysah MM, Siatskas C, Petratos S. Hematopoietic stem cell transplantation for multiple sclerosis: Is it a clinical reality? Stem Cell Res Ther. 2016;7:12.
9. Barbado J, Tabera S, Sánchez A, García-Sancho J. Therapeutic potential of allogeneic mesenchymal stromal cells transplantation for lupus nephritis. Lupus. 2018;27(13):2161-2165.
10. Batsali AK, Kastrinaki MC, Papadaki HA, Pontikoglou C. Mesenchymal stem cells derived from Wharton's Jelly of the umbilical cord: Biological properties and emerging clinical applications. Curr

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Stem Cell Res Ther. 2013;8(2):144-155.
11. Bauchat A, Polishchuk V, Fabrizio VA, et al. Safety and feasibility of umbilical cord blood transplantation in children with neuronal ceroid lipofuscinosis: A retrospective study. Stem Cells Transl Med. 2025;14(1):szae080.
12. Beland B, Storek J, Quartermain L, et al. Refractory myasthenia gravis treated with autologous hematopoietic stem cell transplantation. Clin Transl Neurol. 2025;12(1):56-68.
13. Berry JD, Cudkowicz ME, Windebank AJ, et al. NurOwn, phase 2, randomized, clinical trial in patients with ALS: Safety, clinical, and biomarker results. Neurology. 2019;93(24):e2294-e2305.
14. Bingham SJ, Snowden JA, Emery P. Autologous blood stem cell transplantation as therapy for autoimmune diseases. Ann Med. 2000;32(9):615-621.
15. Binks M, Passweg JR, Furst D, et al. Phase I/II trial of autologous stem cell transplantation in systemic sclerosis: Procedure related mortality and impact on skin disease. Ann Rheum Dis. 2001;60(6):577-584.
16. BlueCross BlueShield Association (BCBSA), Technology Evaluation Center (TEC). High-dose lymphoablative therapy (HDLT) with or without stem-cell rescue for treatment of severe autoimmune diseases. TEC Assessment Program. Chicago, IL: BCBSA; November 2001;16(14).
17. Boffa G, Signori A, Massacesi L, et al; Italian BMT-MS Study Group and the Italian MS Register. Hematopoietic stem cell transplantation in people with active secondary progressive multiple sclerosis. Neurology. 2023;100(11):e1109-e1122.
18. Bousvaros A, Leichtner A, Burpee T. Treatment of ulcerative colitis in children and adolescents. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed May 2013.
19. Bredeson CN, Pavletic SZ. Considerations when designing a clinical trial of haematopoietic stem cell transplantation for autoimmune disease. Best Pract Res Clin Haematol. 2004;17(2):327-343.
20. Brierley CK, Castilla-Llorente C, Labopin M, et al; European Society for Blood and Marrow Transplantation (EBMT) Autoimmune Diseases Working Party (ADWP). Autologous haematopoietic stem cell transplantation for Crohn's disease: A retrospective survey of long-term outcomes from the European Society for Blood and Marrow Transplantation. J Crohns Colitis. 2018;12(9):1097-1103.
21. Brodsky RA, Petri M, Jones RJ. Hematopoietic stem cell transplantation for systemic lupus erythematosus. Rheum Dis Clin North Am. 2000;26(2):377-387, viii.
22. Burman J, Kirgizov K, Carlson K, et al. Autologous hematopoietic stem cell transplantation for pediatric multiple sclerosis: A registry-based study of the Autoimmune Diseases Working Party (ADWP) and Pediatric Diseases Working Party (PDWP) of the European Society for Blood and Marrow Transplantation (EBMT). Bone Marrow Transplant. 2017;52(8):1133-1137.
23. Burt RK, Balabanov R, Burman J, et al. Effect of nonmyeloablative hematopoietic stem cell transplantation vs continued disease-modifying therapy on disease progression in patients with relapsing-remitting multiple sclerosis: A randomized clinical trial. JAMA. 2019;321(2):165-174.
24. Burt RK, Balabanov R, Han X, et al. Association of nonmyeloablative hematopoietic stem cell transplantation with neurological disability in patients with relapsing-remitting multiple sclerosis. JAMA. 2015;313(3):275-284.
25. Burt RK, Balabanov R, Han X, et al. Autologous hematopoietic stem cell transplantation for stiff-person spectrum disorder: A clinical trial. Neurology. 2021;96(6):e817-e830.
26. Burt RK, Balabanov R, Han X, et al. Autologous nonmyeloablative hematopoietic stem cell transplantation for neuromyelitis optica. Neurology. 2019;93(18):e1732-e1741.
27. Burt RK, Loh Y, Cohen B, et al. Autologous non-myeloablative haemopoietic stem cell transplantation in relapsing-remitting multiple sclerosis: A phase I/II study. Lancet Neurol. 2009;8(3):244-253.
28. Burt RK, Loh Y, Pearce W, et al. Clinical applications of blood-derived and marrow-derived stem cells for nonmalignant diseases. JAMA. 2008;299(8):925-936.
29. Burt RK, Shah SJ, Dill K, et al. Autologous non-myeloablative haemopoietic stem-cell transplantation compared with pulse cyclophosphamide once per month for systemic sclerosis (ASSIST): An open-label, randomised phase 2 trial. Lancet. 2011;378(9790):498-506.
30. Burt RK, Traynor A, Burns W. Hematopoietic stem cell transplantation of multiple sclerosis, rheumatoid arthritis, and systemic lupus erythematosus. Cancer Treat Res. 1999;101:157-184.
31. Burt RK, Traynor A, Statkute L, et al. Nonmyeloablative hematopoietic stem cell transplantation for systemic lupus erythematosus. JAMA. 2006;295(5):527-535.
32. Burt RK, Traynor A. Hematopoietic stem cell therapy of autoimmune diseases. Curr Opin Hematol. 1998;5(6):472-477.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

33. Burt RK, Traynor AE, Pope R, et al. Treatment of autoimmune disease by intense immunosuppressive conditioning and autologous hematopoietic stem cell transplantation. Blood. 1998;92(10):3505-3514.
34. Burt RK, Traynor AE. SLE - hematopoietic stem cell transplantation for systemic lupus erythematosus. Arthritis Res Ther. 2003;5(5):207-109.
35. Cario H, McMullin MF, Pahl HL. Clinical and hematological presentation of children and adolescents with polycythemia vera. Ann Hematol. 2009;88(8):713-719.
36. Carpenter KLH, Major S, Tallman C, et al. White matter tract changes associated with clinical improvement in an open-label trial assessing autologous umbilical cord blood for treatment of young children with autism. Stem Cells Transl Med. 2019;8(2):138-147.
37. Casanova B, Jarque I, Gascón F, et al. Autologous hematopoietic stem cell transplantation in relapsing-remitting multiple sclerosis: Comparison with secondary progressive multiple sclerosis. Neurol Sci. 2017;38(7):1213-1221.
38. Chez M, Lepage C, Parise C, et al. Safety and observations from a placebo-controlled, crossover study to assess Use of autologous umbilical cord blood stem cells to improve symptoms in children with autism. Stem Cells Transl Med. 2018;7(4):333-341.
39. Chihaby N, Orliaguet M, Le Pottier L, et al. Treatment of Sjogren's syndrome with mesenchymal stem cells: A systematic review. Int J Mol Sci. 2021;22(19):10474.
40. Cipriani P, Ruscitti P, Giacomelli R. Stem cell therapies for systemic sclerosis. Br J Haematol. 2015;168(3):328-337.
41. Cohen JA, Baldassari LE, Atkins HL, et al. Autologous hematopoietic cell transplantation for treatment-refractory relapsing multiple sclerosis: Position statement from the American Society for Blood and Marrow Transplantation. Biol Blood Marrow Transplant. 2019;25(5):845-854.
42. Comi G, Kappos L, Clanet M, et al. Guidelines for autologous blood and marrow stem cell transplantation in multiple sclerosis: A consensus report written on behalf of the European Group for Blood and Marrow Transplantation and the European Charcot Foundation. BMT-MS Study Group. J Neurol. 2000;247(5):376-382.
43. Daikeler T, Hügle T, Farge D, et al; Working Party Autoimmune Diseases of the EBMT. Allogeneic hematopoietic SCT for patients with autoimmune diseases. Bone Marrow Transplant. 2009;44(1):27-33.
44. Dawson G, Sun JM, Baker J, et al. A phase II randomized clinical trial of the safety and efficacy of intravenous umbilical cord blood infusion for treatment of children with autism spectrum disorder. J Pediatr. 2020;222:164-173.
45. Dawson G, Sun JM, Davlantis KS, et al. Autologous cord blood infusions are safe and feasible in young children with autism spectrum disorder: Results of a single-center phase I open-label trial. Stem Cells Transl Med. 2017;6(5):1332-1339.
46. Del Papa N, Onida F, Zaccara E, et al. Autologous hematopoietic stem cell transplantation has better outcomes than conventional therapies in patients with rapidly progressive systemic sclerosis. Bone Marrow Transplant. 2017;52(1):53-58.
47. Du H, Lim SL, Grob S, Zhang K. Induced pluripotent stem cell therapies for geographic atrophy of age-related macular degeneration. Semin Ophthalmol. 2011;26(3):216-224.
48. El-Badawy A, El-Badri N. Clinical efficacy of stem cell therapy for diabetes mellitus: A meta-analysis. PLoS One. 2016;11(4):e0151938.
49. Factor SA, Esper CD. Multiple system atrophy: Prognosis and treatment. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed April 2020.
50. Falk RJ, Dall'Era M, Appel GB. Treatment and prognosis of diffuse or focal proliferative lupus nephritis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed April 2019.
51. Fassas A, Anagnostopoulos A, Kazis A, et al. Autologous stem cell transplantation in progressive multiple sclerosis -- an interim analysis of efficacy. J Clin Immunol. 2000;20(1):24-30.
52. Fassas A, Kazis A. High-dose immunosuppression and autologous hematopoietic stem cell rescue for severe multiple sclerosis. J Hematother Stem Cell Res. 2003;12(6):701-711.
53. Fassas A, Kimiskidis VK. Stem cell transplantation for multiple sclerosis: What is the evidence? Blood Rev. 2003 Dec;17(4):233-240.
54. Furst DE. Rational therapy in the treatment of systemic sclerosis. Curr Opin Rheumatol. 2000;12(6):540-544.
55. Furst DE. Stem cell transplantation for autoimmune disease: Progress and problems. Curr Opin Rheumatol. 2002;14(3):220-224.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

56. Furst DE. The status of stem cell transplantation for rheumatoid arthritis: A rheumatologist's view. J Rheumatol Suppl. 2001;64:60-61.
57. Glass JD, Boulis NM, Johe K, et al. Lumbar intraspinal injection of neural stem cells in patients with ALS: Results of a phase I trial in 12 patients. Stem Cells. 2012;30(6):1144-1151.
58. Glass JD, Hertzberg VS, Boulis NM, et al. Transplantation of spinal cord-derived neural stem cells for ALS: Analysis of phase 1 and 2 trials. Neurology. 2016;87(4):392-400.
59. Gratwohl A, Passweg J, Bocelli-Tyndall C, et al. Autologous hematopoietic stem cell transplantation for autoimmune diseases. Bone Marrow Transplant. 2005;35(9):869-879.
60. Griffith LM, Pavletic SZ, Tyndall A, et al. Feasibility of allogeneic hematopoietic stem cell transplantation for autoimmune disease: Position statement from a National Institute of Allergy and Infectious Diseases and National Cancer Institute-Sponsored International Workshop, Bethesda, MD, March 12 and 13, 2005. Biol Blood Marrow Transplant. 2005;11(11):862-870.
61. Grigg A, Tubridy NJ, Szer J, et al. Cladribine followed by autologous stem-cell transplantation in progressive multiple sclerosis. Intern Med J. 2004;34(1-2):66-69.
62. Hauser SL. Hematopoietic stem cell transplantation for MS: Extraordinary evidence still needed. JAMA. 2015;313(3):251-252.
63. Hawkey CJ, Allez M, Clark MM, et al. Autologous hematopoetic stem cell transplantation for refractory Crohn disease: A randomized clinical trial. JAMA. 2015;314(23):2524-2534.
64. He J, Kong D, Yang Z, et al. Clinical efficacy on glycemic control and safety of mesenchymal stem cells in patients with diabetes mellitus: Systematic review and meta-analysis of RCT data. PLoS One. 2021;16(3):e0247662.
65. Hendrawan K, Khoo MLM, Visweswaran M, et al. Long-term suppression of circulating proinflammatory cytokines in multiple sclerosis patients following autologous haematopoietic stem cell transplantation. Front Immunol. 2022;12:782935.
66. Hernanz N, Sierra M, Volpato N, et al. Autologous haematopoietic stem cell transplantation in refractory Crohn's disease: Experience in our centre. Gastroenterol Hepatol. 2019;42(1):16-22.
67. Hintzen RQ. Stem cell transplantation in multiple sclerosis: Multiple choices and multiple challenges. Mult Scler. 2002;8(2):155-160.
68. Host L, Nikpour M, Calderone A, et al. Autologous stem cell transplantation in systemic sclerosis: A systematic review. Clin Exp Rheumatol. 2017;35 Suppl 106(4):198-207.
69. Huang Y, Enzmann V, Ildstad ST. Stem cell-based therapeutic applications in retinal degenerative diseases. Stem Cell Rev. 2011;7(2):434-445.
70. Inglese M, Mancardi GL, Pagani E, et al. and the Italian GITMO-NEURO Group on Autologous Hematopoietic Stem Cell Transplantation. Brain tissue loss occurs after suppression of enhancement in patients with multiple sclerosis treated with autologous haematopoietic stem cell transplantation. J Neurol Neurosurg Psychiatry. 2004;75(4):643-644.
71. Jacobson DL, Gange SJ, Rose NR, et al. Epidemiology and estimated population burden of selected autoimmune diseases in the United States. Clin Immunol Immunopathol. 1997;84(3):223-243.
72. Jaime-Perez JC, Melendez-Flores JD, Ramos-Davila EM, et al. Hematopoietic stem cell transplantation for uncommon immune-mediated neurological disorders: A literature review. Cytotherapy. 2022 Jul;24(7):676-685.
73. Jantunen E, Luosujarvi R. Stem cell transplantation in autoimmune diseases: An update. Ann Med. 2005;37(7):533-541.
74. Jantunen E, Myllykangas-Luosujarvi R. Stem cell transplantation for treatment of severe autoimmune diseases: Current status and future perspectives. Bone Marrow Transplant. 2000;25(4):351-356.
75. Jayne D, Passweg J, Marmont A, et al., and the European Group for Blood and Marrow Transplantation; European League Against Rheumatism Registry. Autologous stem cell transplantation for systemic lupus erythematosus. Lupus. 2004;13(3):168-176.
76. Kaplan AA, George JN. Treatment and prognosis of thrombotic thrombocytopenic purpura-hemolytic uremic syndromes in adults. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed May 2013.
77. Kimura Y, Schulert G. Systemic juvenile idiopathic arthritis: Treatment and prognosis. UpToDate [online serial], Waltham, MA: UpToDate; reviewed May 2025.
78. Kishimoto T, Hamazaki T, Yasui M, et al. Autologous hematopoietic stem cell transplantation for 3 patients with severe juvenile rheumatoid arthritis. Int J Hematol. 2003;78(5):453-456.
79. Klippel JH. Systemic lupus erythematosus: Demographics, prognosis, and outcome. J Rheumatol Suppl. 1997;48:67-71.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

80. Kotlarz D, Beier R, Murugan D, et al. Loss of interleukin-10 signaling and infantile inflammatory bowel disease: Implications for diagnosis and therapy. Gastroenterology. 2012;143(2):347-355.
81. Kotter I, Daikeler T, Amberger C, et al. Autologous stem cell transplantation of treatment-resistant systemic vasculitis--a single center experience and review of the literature. Clin Nephrol. 2005;64(6):485-489.
82. Kowal-Bielecka O, Fransen J, Avouac J, et al; EUSTAR Coauthors. Update of EULAR recommendations for the treatment of systemic sclerosis. Ann Rheum Dis. 2017;76(8):1327-1339.
83. Kozak T, Havrdova E, Pit'ha J, et al. Immunoablative therapy with autologous stem cell transplantation in the treatment of poor risk multiple sclerosis. Transplant Proc. 2001;33(3):2179-2181.
84. Kozak T, Rychlik I. Developments in hematopoietic stem-cell transplantation in the treatment of autoimmune diseases. Isr Med Assoc J. 2002;4(4):268-271.
85. Lee PH, Lee JE, Kim HS, et al. A randomized trial of mesenchymal stem cells in multiple system atrophy. Ann Neurol. 2012;72(1):32-40.
86. Lei R, Liu X, Wu Y, et al. Application of allogeneic hematopoietic stem cell transplantation to treat Behcet's disease with myelodysplastic syndrome: A case report and literature review. Ann Hematol. 2024;103(12):6015-6020.
87. Leone A, Radin M, Almarzooqi AM, et al. Autologous hematopoietic stem cell transplantation in systemic lupus erythematosus and antiphospholipid syndrome: A systematic review. Autoimmun Rev. 2017;16(5):469-477.
88. Leung Y, Geddes M, Storek J, et al. Hematopoietic cell transplantation for Crohn's disease; is it time? World J Gastroenterol. 2006;12(41):6665-6673.
89. Li C, Feng J, Chen S, He Y. Efficacy of hematopoietic stem cell for multiple sclerosis, an evidence based meta-analysis. Cell Mol Biol (Noisy-le-grand). 2016;62(4):48-52.
90. Li Y, Tsai YT, Hsu CW, et al. Long-term safety and efficacy of human-induced pluripotent stem cell (iPS) grafts in a preclinical model of retinitis pigmentosa. Mol Med. 2012;18:1312-1319.
91. Liang J, Zhang H, Zhao C, et al. Effects of allogeneic mesenchymal stem cell transplantation in the treatment of liver cirrhosis caused by autoimmune diseases. Int J Rheum Dis. 2017;20(9):1219-1226.
92. Lira-Lara O, Robles-Nasta M, Olivares-Gazca JC, et al. Early morbimortality in autologous hematopoietic cell transplantation performed on outpatient basis in patients with autoimmune diseases: Experience in 1700 patients. Bone Marrow Transplant. 2025;60(5):640-644.
93. Lisukov IA, Sizikova SA, Kulagin AD, et al. High-dose immunosuppression with autologous stem cell transplantation in severe refractory systemic lupus erythematosus. Lupus. 2004;13(2):89-94.
94. Liu B, Shu S, Kenny TP, et al. Stem cell therapy in autoimmune rheumatic diseases: A comprehensive review. Clin Rev Allergy Immunol. 2014;47(2):244-257.
95. Liu J, Yang Y, Qi Y. Efficacy of mesenchymal stromal cells in the treatment of type 1 diabetes: A systematic review. Tissue Bank. 2024;25(2):663-676.
96. Loh Y, Oyama Y, Statkute L, et al. Autologous hematopoietic stem cell transplantation in systemic lupus erythematosus patients with cardiac dysfunction: Feasibility and reversibility of ventricular and valvular dysfunction with transplant-induced remission. Bone Marrow Transplant. 2007;40(1):47-53.
97. Lowenthal RM, Graham SR. Does hemopoietic stem cell transplantation have a role in treatment of severe rheumatoid arthritis? J Clin Immunol. 2000;20(1):17-23.
98. Madani S, Amanzadi M, Aghayan HR, et al. Investigating the safety and efficacy of hematopoietic and mesenchymal stem cell transplantation for treatment of T1DM: A systematic review and meta-analysis. Syst Rev. 2022 ;11(1):82.
99. Mancardi GL, Saccardi R, Filippi M, et al. Autologous hematopoietic stem cell transplantation suppresses Gd-enhanced MRI activity in MS. Neurology. 2001;57(1):62-68.
100. Mancardi GL, Sormani MP, Gualandi F; ASTIMS Haemato-Neurological Collaborative Group, On behalf of the Autoimmune Disease Working Party (ADWP) of the European Group for Blood and Marrow Transplantation (EBMT); ASTIMS Haemato-Neurological Collaborative Group On behalf of the Autoimmune Disease Working Party ADWP of the European Group for Blood and Marrow Transplantation EBMT. Autologous hematopoietic stem cell transplantation in multiple sclerosis: A phase II trial. Neurology. 2015;84(10):981-988.
101. Mariottini A, Bulgarini G, Cornacchini S, et al. Hematopoietic stem cell transplantation for the treatment of autoimmune neurological diseases: An update. Bioengineering (Basel). 2023;10(2):176.
102. Marmont AM. Hematopoietic stem cell transplantation for severe and refractory lupus: comment on the article by Traynor et al. Arthritis Rheum. 2003;48(9):2696-2697;

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

103. Marmont AM. New horizons in the treatment of autoimmune diseases: Immunoablation and stem cell transplantation. Annu Rev Med. 2000;51:115-134.
104. Marmont AM. Will hematopoietic stem cell transplantation cure human autoimmune diseases? J Autoimmun. 2008;30(3):145-150.
105. Martinez HR, Molina-Lopez JF, Alez-Garza MT, et al. Stem cell transplantation in amyotrophic lateral sclerosis patients. Methodological approach, safety, and feasibility. Cell Transplant. 2012;21(9):1899-1907.
106. Matt MG, Drozdov D, Bendstrup E, et al. Allogeneic haematopoietic stem-cell transplantation for children with refractory systemic juvenile idiopathic arthritis and associated lung disease: Outcomes from an international, retrospective cohort study. Lancet Rheumatol. 2025;7(4):e243-e251.
107. Matthews JH, Smith CA, Herst J, et al.; Hematology Disease Site Group. The management of malignant thrombocytosis in Philadelphia chromosome-negative myeloproliferative disease: Guideline recommendations. Toronto, ON: Cancer Care Ontario (CCO); January 15, 2008.
108. M'Barek KB , Habeler W, Regent F, Monville C. Developing cell-based therapies for RPE-associated degenerative eye diseases. Adv Exp Med Biol. 2019;1186:55-97.
109. Medina R, Derias A-M, Lakdawala M, et al. Overview of emerging therapies for demyelinating diseases. World J Clin Cases. 2024;12(30):6361-6373.
110. Mobasheri A, Csaki C, Clutterbuck AL, et al. Mesenchymal stem cells in connective tissue engineering and regenerative medicine: Applications in cartilage repair and osteoarthritis therapy. Histol Histopathol. 2009;24(3):347-366.
111. Muise AM, Snapper SB, Kugathasan S. The age of gene discovery in very early onset inflammatory bowel disease. Gastroenterology. 2012; 143(2):285-288.
112. Mundy L, Hiller J. Autologous haematopoietic stem cell transplantation (HSCT) for the regeneration of insulin-producing pancreatic cells in type 1 diabetics. Horizon Scanning Prioritising Summary. Adelaide, SA: Adelaide Health Technology Assessment; March 2010.
113. Murano PA, Pasquini M, Atkins HL, et al; for the Multiple Sclerosis-Autologous Hematopoietic Stem Cell Transplantation (MS-AHSCT) Long-term Outcomes Study Group. Long-term outcomes after autologous hematopoietic stem cell transplantation for multiple sclerosis. JAMA Neurol. 2017;74(4):459-469..
114. Nabizadeh F, Masrouri S, Sharifkazemi H, et al. Autologous hematopoietic stem cell transplantation in neuromyelitis optica spectrum disorder: A systematic review and meta-analysis. J Clin Neurosci. 2022;105:37-44.
115. Nair V, Vasdev V, Kumar A, et al. Stem cell transplant in systemic sclerosis: An Indian experience. Int J Rheum Dis. 2018;21(4):859-865.
116. Nash RA, Dansey R, Storek J, et al. Epstein-Barr virus-associated posttransplantation lymphoproliferative disorder after high-dose immunosuppressive therapy and autologous CD34-selected hematopoietic stem cell transplantation for severe autoimmune diseases. Biol Blood Marrow Transplant. 2003;9(9):583-591.
117. Nash RA, Hutton GJ, Racke MK, et al. High-dose immunosuppressive therapy and autologous HCT for relapsing-remitting MS. Neurology. 2017;88(9):842-852.
118. Nash RA. Allogeneic HSCT for autoimmune diseases: Conventional conditioning regimens. Bone Marrow Transplant. 2003;32 Suppl 1:S77-S80.
119. Nash RA. Prospects of stem cell transplantation in autoimmune diseases. J Clin Immunol. 2000;20(1):38-45.
120. National Health, Lung and Blood Institute (NHLBI). Polycythemia vera. Diseases and Conditions Index. Bethesda, MD: National Institutes for Health (NIH); revised March 2011. Available at: http://www.nhlbi.nih.gov/health/dci/Diseases/poly/poly_treatments.html. Accessed July 12, 2011.
121. National Institute for Health and Clinical Excellence (NICE). Crohn's disease: Management in adults, children and young people. NICE Clinical Guideline No. 152. London, UK: National Institute for Health and Clinical Excellence (NICE); October 2012.
122. Nicholas RS, Rhone EE, Mariottini A, et al; London Group on Autologous Hematopoietic Stem Cell Transplantation for Multiple Sclerosis. Autologous hematopoietic stem cell transplantation in active multiple sclerosis: A real-world case series. Neurology. 2021;97(9):e890-e901.
123. Nikolov NP, Pavletic SZ. Technology Insight: Hematopoietic stem cell transplantation for systemic rheumatic disease. Nat Clin Pract Rheumatol. 2008;4(4):184-191.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

124. Olek MJ, Mowry E. Overview of disease-modifying therapies for multiple sclerosis. UpToDate [online serial], Waltham, MA: UpToDate; reviewed May 2025.
125. Oyama Y, Traynor AE, Barr W, Burt RK. Allogeneic stem cell transplantation for autoimmune diseases: Nonmyeloablative conditioning regimens. Bone Marrow Transplant. 2003;32 Suppl 1:S81-S83.
126. Palma CA, Lindeman R, Tuch BE. Blood into beta-cells: Can adult stem cells be used as a therapy for Type 1 diabetes? Regen Med. 2008;3(1):33-47.
127. Palma JA, Kaufmann H. Novel therapeutic approaches in multiple system atrophy. Clin Auton Res. 2015;25(1):37-45.
128. Passweg JR, Rabusin M, Musso M, et al. and the Autoimmune Disease Working Party of the EBMT. Haematopoetic stem cell transplantation for refractory autoimmune cytopenia. Br J Haematol. 2004;125(6):749-755.
129. Patti F, Chisari CG, Toscano S, et al. Autologous hematopoietic stem cell transplantation in multiple sclerosis patients: Monocentric case series and systematic review of the literature. J Clin Med. 2022;11(4):942.
130. Peppercorn MA, Farrell RJ. Management of severe ulcerative colitis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed May 2013.
131. Pluchino S, Martino G. The therapeutic use of stem cells for myelin repair in autoimmune demyelinating disorders. J Neurol Sci. 2005;233(1-2):117-119.
132. Press R, Askmark H, Svenningsson A, et al. Autologous haematopoietic stem cell transplantation: A viable treatment option for CIDP. J Neurol Neurosurg Psychiatry. 2014;85(6):618-624.
133. Raj A, Bertolone S, Cheerva A. Successful treatment of refractory autoimmune hemolytic anemia with monthly rituximab following nonmyeloablative stem cell transplantation for sickle cell disease. J Pediatr Hematol Oncol. 2004;26(5):312-314.
134. Reston JT, Uhl S, Treadwell JR, et al. Autologous hematopoietic cell transplantation for multiple sclerosis: A systematic review. Multiple Sclerosis. 2011;17(2):204-213.
135. Rice CM, Kemp K, Wilkins A, Scolding NJ. Cell therapy for multiple sclerosis: An evolving concept with implications for other neurodegenerative diseases. Lancet. 2013;382(9899):1204-1213.
136. Riller Q, Hauri-Hohl MM, Lum SH, et al. HLA-haploidentical stem cell transplantation for chronic granulomatous disease: An EBMT-IEWP retrospective study. Blood. 2025;145(21):2518-2527.
137. Saccardi R, Kozak T, Bocelli-Tyndall C, et al; Autoimmune Diseases Working Party of EBMT. Autologous stem cell transplantation for progressive multiple sclerosis: Update of the European Group for Blood and Marrow Transplantation autoimmune diseases working party database. Mult Scler. 2006;12(6):814-823.
138. Sattwika PD, Mustafa R, Paramaiswari A, Herningtyas EH. Stem cells for lupus nephritis: A concise review of current knowledge. Lupus. 2018;27(12):1881-1897.
139. Schippling S, Martin R. Stem cell therapy in multiple sclerosis: A clinical update. Z Rheumatol. 2009;68(3):214-215, 217-219.
140. Schwartz SD, Hubschman JP, Heilwell G, et al. Embryonic stem cell trials for macular degeneration: A preliminary report. Lancet. 2012;379(9817):713-720.
141. Seissler J, Schott M. Generation of insulin-producing beta cells from stem cells--perspectives for cell therapy in type 1 diabetes. Horm Metab Res. 2008;40(2):155-161.
142. Singh RK, Binette F, Seiler M, et al. Pluripotent stem cell-based organoid technologies for developing next-generation vision restoration therapies of blindness. J Ocul Pharmacol Ther. 2021;37(3):147-156.
143. Singer W, Dietz AB, Zeller AD, et al. Intrathecal administration of autologous mesenchymal stem cells in multiple system atrophy. Neurology. 2019;93(1):e77-e87
144. Snowden JA, Biggs JC, Milliken ST, et al. A phase I/II dose escalation study of intensified cyclophosphamide and autologous blood stem cell rescue in severe, active rheumatoid arthritis. Arthritis Rheum. 1999;42(11):2286-2292.
145. Snowden JA, Brooks PM, Biggs JC. Haemopoietic stem cell transplantation for autoimmune diseases. Br J Haematol. 1997;99(1):9-22.
146. Snowden JA, Passweg J, Moore JJ, et al. Autologous hemopoietic stem cell transplantation in severe rheumatoid arthritis: A report from the EBMT and ABMTR. J Rheumatol. 2004;31(3):482-488.
147. Snowden JA, Saccardi R, Allez M, et al. Haematopoietic SCT in severe autoimmune diseases: Updated guidelines of the European Group for Blood and Marrow Transplantation. Bone Marrow Transplant. 2012;47(6):770-790.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

148. Song XN, Lv HY, Sun LX, et al. Autologous stem cell transplantation for systemic lupus erythematosus: Report of efficacy and safety at 7 years of follow-up in 17 patients. Transplant Proc. 2011;43(5):1924-1927.
149. Sormani MP, Muraro PA, Schiavetti I, et al. Autologous hematopoietic stem cell transplantation in multiple sclerosis: A meta-analysis. Neurology. 2017;88(22):2115-2122.
150. St Clair EW, McCallum R. Cogan syndrome. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed May 2023.
151. Stellmann JP, Sturner KH, Ufer F, et al. Stem cell transplantation for multiple sclerosis. Hamburg experiences and state of international research. Nervenarzt. 2015;86(8):989-996.
152. Sullivan KM, Goldmuntz EA, Keyes-Elstein L, et al; SCOT Study Investigators. Myeloablative autologous stem-cell transplantation for severe scleroderma. N Engl J Med. 2018;378(1):35-47.
153. Sykes M, Nikolic B. Treatment of severe autoimmune disease by stem-cell transplantation. Nature. 2005;435(7042):620-627.
154. Tappenden P, Saccardi R, Confavreux C, et al. Autologous haematopoietic stem cell transplantation for secondary progressive multiple sclerosis: An exploratory cost-effectiveness analysis. Bone Marrow Transplant. 2010;45(6):1014-1021.
155. Tefferi A, Vainchenker W. Myeloproliferative neoplasms: Molecular pathophysiology, essential clinical understanding, and treatment strategies. J Clin Oncol. 2011;29(5):573-582.
156. Thomas ED. Pros and cons of stem cell transplantation for autoimmune disease. J Rheumatol Suppl. 1997;48:100-102.
157. Thomsen GM, Gowing G, Svendsen S, Svendsen CN. The past, present and future of stem cell clinical trials for ALS. Exp Neurol. 2014;262 Pt B:127-137.
158. Traynor AE, Schroeder J, Rosa RM, et al. Treatment of severe systemic lupus erythematosus with high-dose chemotherapy and haemopoietic stem-cell transplantation: A phase I study. Lancet. 2000;356(9231):701-707.
159. Tyndall A, Fassas A, Passweg J, et al. Autologous haematopoietic stem cell transplants for autoimmune disease -- feasibility and transplant-related mortality. Autoimmune Disease and Lymphoma Working Parties of the European Group for Blood and Marrow Transplantation, the European League Against Rheumatism and the International Stem Cell Project for Autoimmune Disease. Bone Marrow Transplant. 1999;24(7):729-734.
160. Tyndall A, Matucci-Cerinic M. Haematopoietic stem cell transplantation for the treatment of systemic sclerosis and other autoimmune disorders. Expert Opin Biol Ther. 2003;3(7):1041-1049.
161. Tyndall A. Immunoablation and haemopoietic stem cell transplantation for severe autoimmune disease with special reference to systemic lupus erythematosus. Lupus. 2001;10(3):214-215.
162. U.S. Office of Veteran's Affairs, Office of Patient Care Services, Technology Assessment Unit. Bone marrow transplantation for treatment of multiple sclerosis. Bibliography. Boston, MA: Technology Assessment Unit, Office of Patient Care Services, US Department of Veterans Affairs (VATAP); 2006.
163. van Laar JM, Farge D, Tyndall A. Stem cell transplantation: A treatment option for severe systemic sclerosis? Ann Rheum Dis. 2008;67 Suppl 3:iii35-iii38.
164. Verburg RJ, Sont JK, Vliet Vlieland TP, et al. High dose chemotherapy followed by autologous peripheral blood stem cell transplantation or conventional pharmacological treatment for refractory rheumatoid arthritis? A Markov decision analysis. J Rheumatol. 2001;28(4):719-727.
165. Voltarelli JC, Couri CE, Stracieri AB, et al. Autologous nonmyeloablative hematopoietic stem cell transplantation in newly diagnosed type 1 diabetes mellitus. JAMA. 2007;297(14):1568-1576.
166. Voltarelli JC, Ouyang J. Hematopoietic stem cell transplantation for autoimmune diseases in developing countries: Current status and future prospectives. Bone Marrow Transplant. 2003;32 Suppl 1:S69-S71.
167. Wagner JE, Ishida-Yamamoto A, McGrath JA, et al. Bone marrow transplantation for recessive dystrophic epidermolysis bullosa. N Engl J Med. 2010;363(7):629-639.
168. Wang M, Cao C, Sun J, et al. Application of autologous hematopoietic stem cell transplantation for pemphigus. Int J Dermatol. 2017;56(3):296-301.
169. Wang Q, Qian S, Li J, et al. Combined transplantation of autologous hematopoietic stem cells and allogenic mesenchymal stem cells increases T regulatory cells in systemic lupus erythematosus with refractory lupus nephritis and leukopenia. Lupus. 2015;24(11):1221-1226.
170. Weissman L. Autism spectrum disorder in children and adolescents: Overview of management. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed April 2021.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

171. Willison AG, Ruck T, Lenz G, et al. The current standing of autologous haematopoietic stem cell transplantation for the treatment of multiple sclerosis. J Neurol. 2022;269(7):3937-3958.
172. Witkowska M, Smolewska E, Smolewski P. Hematopoietic stem cell transplantation in children with autoimmune connective tissue diseases. Arch Immunol Ther Exp (Warsz). 2014;62(4):319-327.
173. Wong WM, Loh Y, Chan HW, et al. Hematopoietic stem cell transplantation as rescue therapy for refractory autoimmune retinopathy: A case report. Front Immunol. 2025;15:1484798.
174. Wu KH, Wu HP, Chan CK, et al. The role of mesenchymal stem cells in hematopoietic stem cell transplantation: From bench to bedsides. Cell Transplant. 2013;22(4):723-729.
175. Wulffraat NM, Brinkman D, Ferster A, et al, Foster H, Abinun M, Prieur AM, et al. Long-term follow-up of autologous stem cell transplantation for refractory juvenile idiopathic arthritis. Bone Marrow Transplant. 2003;32 Suppl 1:S61-S64.
176. Zeng L, Yu G, Yang K, et al. Efficacy and safety of mesenchymal stem cell transplantation in the treatment of autoimmune diseases (rheumatoid arthritis, systemic lupus erythematosus, inflammatory bowel disease, multiple sclerosis, and ankylosing spondylitis): A systematic review and meta-analysis of randomized controlled trial. Stem Cells Int. 2022;2022:9463314.
177. Zheng Y, Hu J, Sun C, et al. Efficacy of hematopoietic stem cell transplantation treatment in refractory chronic inflammatory demyelinating polyradiculoneuropathy: A systematic review and meta-analysis. Eur J Neurol. 2023;30(8):2570-2582.

Policy History

- Last Review opens in a new browser pop-up window 09/10/2025

  Effective: 04/23/2002

  Next Review: 06/25/2026

- Review History opens in a new browser pop-up window

- Definitions opens in a new browser pop-up window

Additional Information

close popup

- Clinical Policy Bulletin Notes opens in a new browser pop-up window

You are now leaving the Aetna website.

Links to various non-Aetna sites are provided for your convenience only. Aetna Inc. and its subsidiary companies are not responsible or liable for the content, accuracy, or privacy practices of linked sites, or for products or services described on these sites.

Continue opens in new window

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF